Genome-scale transcriptional dynamics and environmental biosensing

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Genome-scale technologies have enabled mapping of the complex molecular networks that govern cellular behavior. An emerging theme in the analyses of these networks is that cells use many layers of regulatory feedback to constantly assess and precisely react to their environment. The importance of complex feedback in controlling the real-time response to external stimuli has led to a need for the next generation of cell-based technologies that enable both the collection and analysis of high-throughput temporal data. Toward this end, we have developed a microfluidic platform capable of monitoring temporal gene expression from over 2,000 promoters. By coupling the "Dynomics" platform with deep neural network (DNN) and associated explainable artificial intelligence (XAI) algorithms, we show how machine learning can be harnessed to assess patterns in transcriptional data on a genome scale and identify which genes contribute to these patterns. Furthermore, we demonstrate the utility of the Dynomics platform as a field-deployable real-time biosensor through prediction of the presence of heavy metals in urban water and mine spill samples, based on the the dynamic transcription profiles of 1,807 unique *Escherichia coli* promoters.

high-throughput microfluidics | dynamics | *E. coli* transcriptomics | explainable AI | biosensor

In model organisms, studying the changing patterns of gene expression in reaction to experimentally induced environmental perturbations often elucidates the underlying signaling network (1–4). However, present techniques to measure genome-wide expression data are often destructive in nature and offer only snapshots of a cell's state (5–11). Meanwhile, an increasing body of evidence points to dynamics as a key way biological systems encode information (12, 13).

Microfluidics, coupled to time-lapse fluorescence microscopy, has served as a means to measure gene-expression dynamics in precisely controlled environments (14–16). Recently, several studies have demonstrated how microfluidic parallelization permits the simultaneous tracking of hundreds to thousands of strains of *Saccharomyces cerevisiae* (17, 18), *Escherichia coli* (19, 20), or mammalian cell lines (21). The combination of microfluidics and genome-scale fluorescent reporter strain libraries has facilitated the study of genomic transcriptional dynamics. Existing approaches, though, have been hampered by short experimental lifespans, limited temporal resolution, static environmental conditions, and the use of single-purpose devices.

In addressing these needs, we have developed Dynomics, a straightforward and broadly applicable research platform that combines multiplexed microfluidics, fluorescence microscopy, and deep neural network (DNN) and explainable artificial intelligence (XAI) algorithms to better resolve transcriptional dynamics at the genome scale. The Dynomics platform enables continuous growth, precise environmental control, and optical monitoring of 2,176 microcolonies of unique GFP-reporter *E. coli* for up to 14 d. In addition to demonstrating the platform's utility in studying transcriptional dynamics through time series and fold-change data, we show the platform's effective application as a continuous biosensor for heavy metals in water supplies at environmentally relevant concentrations and conditions, using DNN algorithms to predict the presence of heavy metals and XAI algorithms to identify which genes contribute to these predictions, unraveling the "black box problem" typical of machine learning (Fig. 1*A*).

Results

Microfluidic Device Development. The Dynomics microfluidic device was designed for straightforward experimental setup, reliable trap filling and cell retention, and optimal fluorescent signal from each spotted microcolony (Fig. 1 *B–D*). The single media inlet–outlet device requires only two fluidic connections after cell spotting and chip bonding. The media inlet channel feeds a total of 2,176 4-μm-tall cell traps. Trap shape and spacing allow a 6,144 Society for Biomolecular Sciences (SBS) density pin pad to deposit cells into the back of the trap, where they grow toward the tapered opening interfacing with 50-μm-tall minor media channels. These minor channels branch off of a larger 230-μm-tall major media channel manifold, which eliminates the possibility of cell trap cross-contamination. Once spotted cells

Significance

New technologies are needed for the global analysis of intracellular signaling networks that encode information in the time domain. We developed a microfluidic platform capable of culturing over 2,000 bacterial strains simultaneously and subjecting them to dynamical perturbation. We used an explainable artificial intelligence classifier to reveal insights embedded in the temporal transcriptional response of *Escherichia coli* exposed to heavy metal stress. This enabled real-time predictions of the presence of heavy metals in complex environmental samples.

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Competing interest statement: W.H.M., M.F., S.C., and J.H. have a financial interest in Quantitative BioSciences. Quantitative BioSciences has an exclusive license to IP stemming from this work, which is owned by the University of California San Diego.

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Data deposition: Preprocessed, labeled machine-learning features, the corresponding library strain position records, and the relevant metadata for our experiments are available on the University of California San Diego Biodynamics Laboratory website (http://biodynamics.ucsd.edu/downloads). The code used to process data from Dynomics experiments and train machine-learning models is available on GitHub at https://github.com/GarrettCGraham/dynomics_public.

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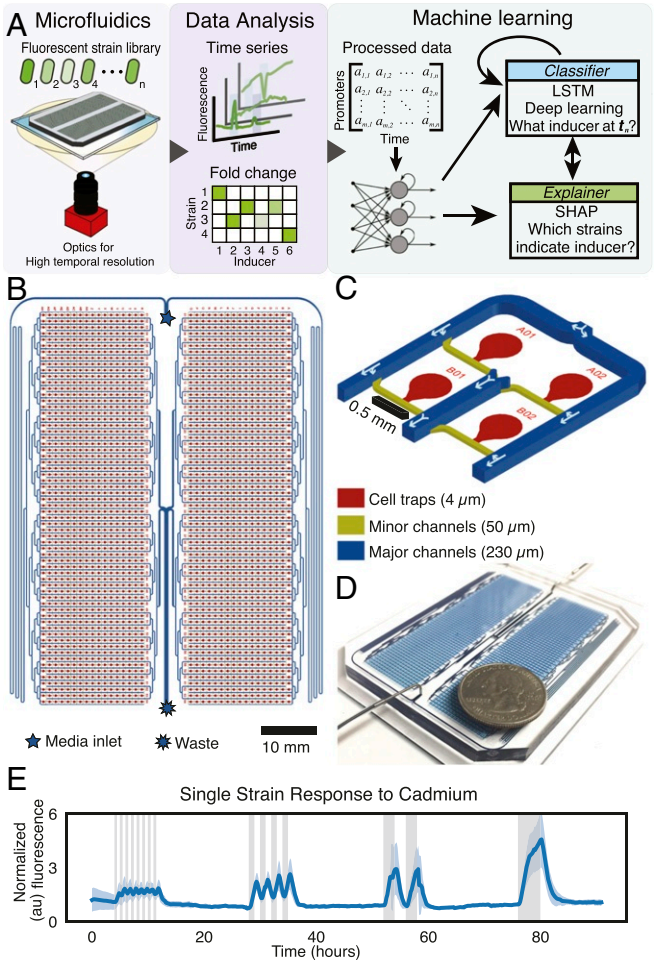

Fig. 1. The Dynomics platform. (*A*) Fluorescent strain libraries are loaded onto large-scale microfluidic devices that can be fully captured in a single image using custom optics. Parallel cultures of *E. coli* are subjected to multiple exposures of different stimuli with time series and fold changes used to quantify responsive strains. Machine-learning algorithms are trained on preprocessed data to enable real-time stimulus detection. (*B*) Design of the Dynomics 2,176-strain microfluidic device with cell traps in red and media channels in blue and yellow. (*C*) Detailed schematic of four cell traps with arrows showing direction of media flow. (*D*) View of the full Dynomics chip. (*E*) Mean fluorescence (solid blue) and SD (shaded blue) of the *E. coli zntA* promoter driving GFP to repeated cadmium inductions (gray bars) with periods increasing from left to right (30 min, 2 h, 4 h, and 8 h).

have reached confluence, inducer compounds can be pulsed in at user-specified frequencies with the dynamic response of each strain measured down to a 4-min temporal resolution (Fig. 1*E*).

Screening for Responsive Promoters to Heavy Metals. Using the Dynomics platform with a previously developed GFP *E. coli* promoter library (22), 1,807 unique *E. coli* promoters were screened against nine heavy metals (Cu(II), Zn(II), Fe(III), Pb(II), Cd(II), Cr(VI), Hg(II), As(III), Sb(III)) at environmentally relevant concentrations (*SI Appendix*, Table S3). Screening experiments lasted 7 to 14 d, with cells exposed to a different heavy metal every 24 h. Promoters responsive to each metal can be identified through a combination of clustering and fold-change analysis. A high-level view of the 1,807 promoter time traces (Fig. 2*A*) and subsequent clustering (Fig. 2*B*) reveals distinct classes of transcriptional responses to a single 4-h zinc exposure. In Fig. 2*B*, clusters 1 and 2 include promoters that are up- and down-regulated, respectively, in the presence of zinc, but return to baseline expression levels within 15 h after zinc removal. Clusters 3 and 4 include promoters that are up- and down-regulated, respectively, but with slower dynamics. Gene ontology (GO) enrichment analysis (*SI Appendix*, Fig. S10) suggests that from these four clusters, genes associated with cellular stress are up-regulated (cellular detoxification, cellular response to toxic substance, and antibiotic catabolic processes) while genes involved in differing metabolic and biosynthesis can be either up- or down-regulated.

Individual responsive strains for each metal were identified, based on their fold-change response (Fig. 2*D*) to daily 4-h metal exposures (Fig. 2*C*). Fold-change measurements highlight the promoters displaying the strongest response to each metal. Subsequent investigation of the most responsive strains (Fig. 2*E*) quantitatively elucidates dynamical properties, such as

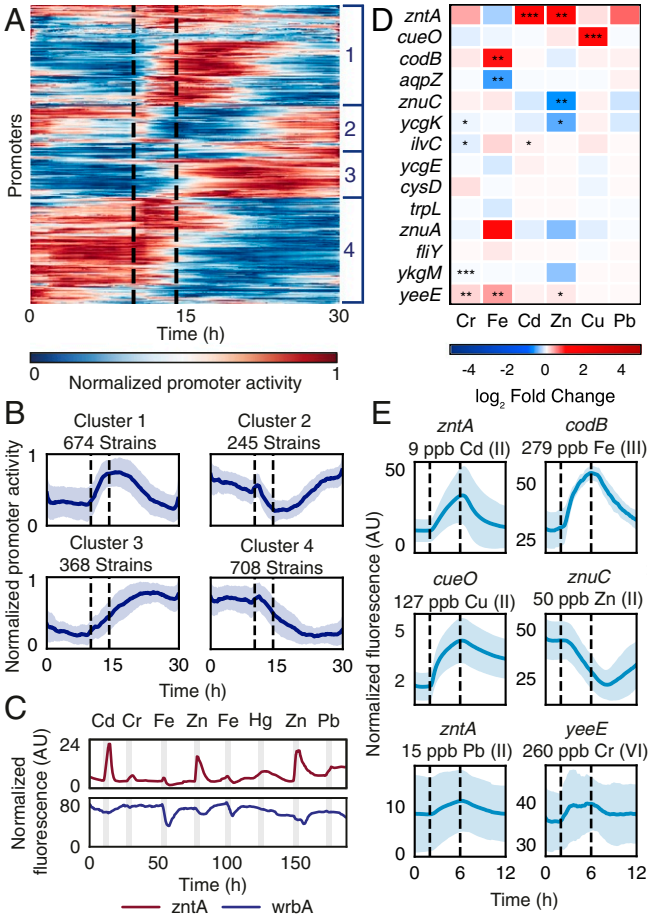

Fig. 2. Dynomics as a screening tool for heavy metal responsive promoters in *E. coli*. (*A*) Fluorescence response of an *E. coli* promoter library during a 4-h 50-ppb Zn induction (dashed window). Each row represents the promoter activity, normalized between 0 and 1, of a single strain, with 1,995 total strains represented. Four clusters from agglomerative clustering are labeled on the right. (*B*) Four clusters of strains calculated from agglomerative clustering from the data in *A*. The mean (dark blue line) ±1 SD (dark blue shading) of all strains in each cluster is plotted. The dashed window denotes when zinc was present. (*C*) Responsive strains over the duration of a Dynomics experiment. Normalized fluorescence for two strains is plotted over the duration of one experiment, with 4-h heavy metal inductions (gray bars) occurring once daily. (*D*) Fold change for top responding strains to all metals. Log_2 of the average fold change is shown for the top responding strains to each heavy metal. $^{*}P = 0.05$, $^{**}P = 0.01$, $^{***}P = 0.001$, respectively. (*E*) Significant single-strain normalized fluorescence response (blue line) ±1 SD (blue shading) across all inductions for a given metal (dashed window).

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amplitude, relaxation time, and response speed, all of which are important factors for their use in the study of gene expression regulation and continuous biosensing applications. While many of the identified sensing strains, such as *zntA* (23) or *cueO* (24), have well-documented metal interactions (*SI Appendix*, Tables S5 and S6), others are less studied or poorly annotated, particularly members of *E. coli* "y-ome" (25). Overall, these analyses demonstrate the utility of this platform as a screening tool for dynamic environmental-response phenotypes in a strain library. However, specific metal discrimination based on fold change alone is difficult to interpret due to promoter nonspecificity, cross-talk, noise, and low-amplitude responses.

Machine Learning. To better discriminate between *E. coli*'s responses to the heavy metals used in our screening, we trained and tested two types of machine-learning models on the Dynomics data (26). The first model, known as extreme gradient boosted trees (XGBoost), is a popular decision tree ensemble-based classifier known for its ability to learn nonlinear models (27). The second one, known as a long short-term memory recurrent neural network (LSTM-RNN), is a DNN (28) selected because of its ability to effectively utilize sample sequence history to classify time series data, a property not shared by XGBoost.

Both classification algorithms outperformed random guessing of the majority class (no toxin) on the standardized experiments' feature set, with the LSTM-RNN performing the best overall (*SI Appendix*, Figs. S8 and S9). As seen by examining the diagonal elements in the confusion matrix in Fig. 3*A*, the LSTM-RNN was able to distinguish both biotic and xenobiotic metal-spiked water from pure water with a high level of reliability.

The LSTM-RNN found iron and copper to be easily detectable biotic metals, which is not surprising given their importance to *E. coli* cellular function (24, 29). Cadmium was the most readily detected xenobiotic metal with the LSTM-RNN classifier, although it was sometimes confused with zinc. *E. coli* are known to use the same sensing and transport systems to capture and export excess amounts of these two metals, which possess the same number of valence electrons (23, 30). Most classification errors occurred during the 10 to 40 min at the start or end of the induction periods, when the LSTM-RNN occasionally had difficulty determining the exact time that each metal was added or removed from the media (Fig. 3*B*). This is most pronounced with the prediction of lead, for which the classifier incorrectly predicted no toxin for 48% of time points where lead was present. This is largely due to the weak promoter responses induced by 0.03 ppm lead, which is only double the Environmental Protection Agency (EPA) maximum contaminant level. In lead exposures with poor prediction, time points at the start of the 4-h induction window are misclassified as no toxin, while lead is accurately predicted near the end of this window (*SI Appendix*, Fig. S10). While past studies have used machine-learning frameworks to assign cells to chronologically distinct phenotypes based on their transcriptomes (31), we believe this is a different instance of a multiclass classifier successfully leveraging genome-wide transcriptional dynamics in live cells to predict exposure of a biological organism to an environmental stressor.

Machine-Learning Introspection Using Explainable Artificial Intelligence. At present, a major obstacle to making scientific conclusions from machine-learning results is the black box problem: As an algorithm's ability to model complex phenomena grows, its decision-making processes become more and more obscured from its operators (32). Recently, explainable artificial intelligence techniques have been employed to explain the decision making of machine-learning algorithms in the life sciences (33–35), while contributions from coalitional game theory have led to the development of a mathematically consistent method for understanding the decision-making process of any AI classifier (36, 37).

Taking advantage of these recent advances, we trained a Shapley additive explanations (SHAP) learner on both our XGBoost and LSTM classifiers (36, 38). The SHAP algorithm scores a strain's impact on the classifier's predictions by calculating Shapley values from cooperative game theory. Shapley values are the mathematically unique way to divide game payout between players who have collaborated with each other to achieve a common goal, assuming basic rules of fairness (39). A major advantage of SHAP is that Lundberg and Lee (36) demonstrated that it is an umbrella method that mathematically unifies several commonly used feature attribution frameworks, including LIME, Layerwise Relevance Propagation, and DeepLIFT. Viewing both SHAP values (impact on classifier output) and feature values (data fed to the classifier) with respect to time offers

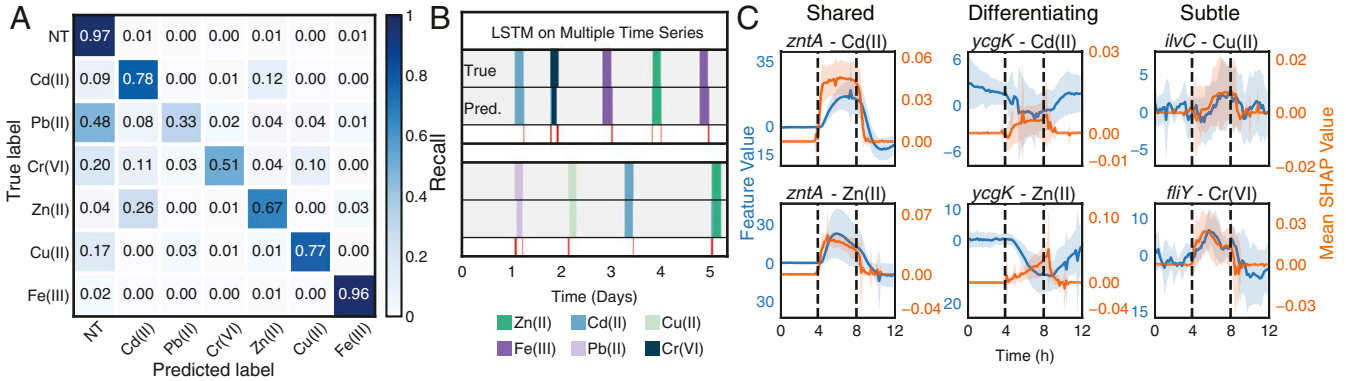

Fig. 3. Machine learning on heavy metal exposures. (*A*) Confusion matrix showing the recall (true positive rate) of the LSTM-RNN classifier in predicting six metals across all experimental data (14,332 time points). (*B*) LSTM-RNN classifier applied to time series data for all six detectable metals in two different experiments. Both experiments have a row for the true media condition and the predicted condition. In the case of correct classification, the color in the predicted row would match the color in the top row, whose color represents which metal was actually present at that time point. An easy way to tell if there has been a misclassification is by seeing if there are any regions flagged with red below the predicted row. Red indicates time points where the prediction does not match the ground truth. (*C*) Feature (blue) and SHAP (orange) time trajectories for individual promoters during metal exposures. Solid lines show the mean value over all inductions for that metal and the shaded region around lines represents SD. Dashed black lines represent metal exposure window. While some promoters are responsive to many different metals, additional information from other promoters helps the classifier to differentiate each metal. Many promoters with noisy and subtle metal responses also contribute to classifier performance.

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insight into how the classifier operates in real time (Fig. 3*C*). The causes of misclassification are made clearer, as SHAP dynamics reveal that the predictive impact of a strain often varies within an induction window, particularly at its start and end. Furthermore, we see how some promoters, such as *zntA*, positively contribute to the detection of multiple metals, which causes the classifier to rely on promoters with less-pronounced responses and lower SHAP value magnitudes to distinguish the exposed metal, explaining some misclassification instances. Finally, promoters that may not have been identified as responsive using fold-change analysis because of subtle, low-amplitude, and noisy responses can be identified via XAI. While these responders may not serve as stand-alone biosensor strains, they provide promising targets for future sensor engineering efforts. These insights highlight the ability of the LSTM-RNN classifier to compile the influence of many strains, prominent and subtle, to make an accurate prediction of the metal exposure.

The SHAP algorithm also highlights similarities and differences between how the LSTM-RNN and XGBoost make decisions. Fig. 4*A* shows the 15 promoters with the highest mean impact on the model, plus the promoterless strain U139, which is included as a negative control. Both methods rely heavily on the metal-sensing promoter *zntA* for the detection and discrimination of multiple metals, especially cadmium and zinc. Beyond *zntA*, XGBoost relies heavily on single strains to detect single metals, in a manner comparable to human attention patterns. The LSTM-RNN, on the other hand, utilizes many strains of moderate influence in a combinatorial fashion; this tendency to find a different representation from that of the human visual system has been noted in other works (40). These trends are also seen when looking at the top 15 promoters for each individual metal class (*SI Appendix*, Fig. S11).

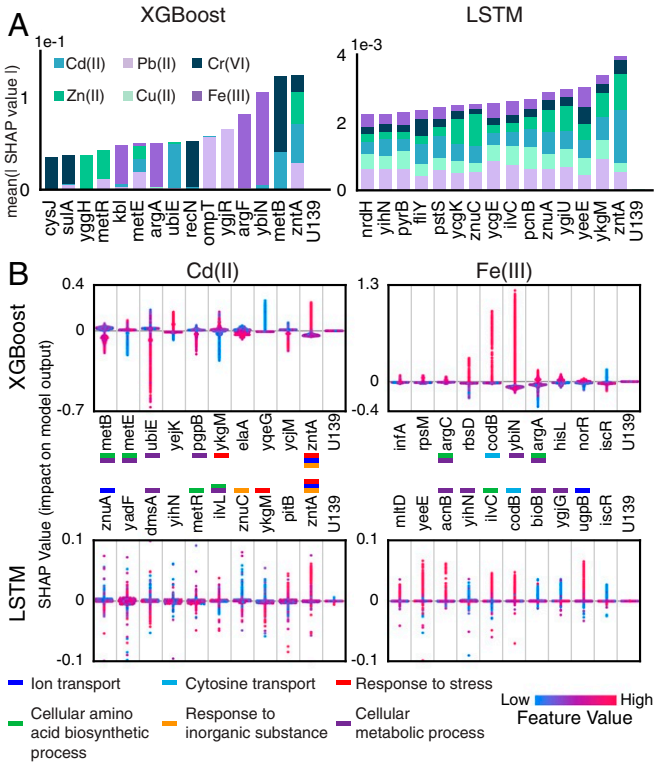

Fig. 4. XAI offers insights into the *E. coli* transcriptional dynamics contributing to metal classification. (*A*) Bar plot showing the cumulative contribution based on the SHAP values of 15 top promoters and a negative control (promoterless strain U139) to the prediction of each metal for both XGBoost and LSTM-RNN classifiers. Colored bars for each metal represent the mean absolute SHAP value over all experimental time points. (*B*) SHAP values shown for 10 top promoters and a negative control (promoterless strain U139) for Cd(II) and Fe(III) for XGBoost and LSTM-RNN. Each point represents the feature value (normalized first derivative) at a given time point. Positive SHAP values suggest that a given metal is present while negative values suggest its absence. Up-regulated promoters (*zntA*, *codB*) give high SHAP values when feature values are high. Promoters are annotated with prominent gene ontology terms enriched between the two datasets.

The ability of the explained classifiers to identify promoters involved in metal response serves as a valuable scientific tool, suggesting potential pathways and genes for further investigation. This value is highlighted by looking at a subset of the 10 most-impactful promoters individually for cadmium and iron inductions (Fig. 4*B*). These summary plots illustrate how the two classifiers make similar decisions through different methods. In the case of cadmium, *zntA* plays a significant role for both classifiers, while different sets of genes involved in ion transport or amino acid synthesis are identified for each. Most notably, the *metE* and *metB* promoters which are involved in methionine synthesis, an amino acid known to chelate cadmium (41), are identified by XGBoost, while the LSTM-RNN uses only the *metE* regulator, *metR*, for detection. Similarly with iron, we see XGBoost rely on members of the arginine synthesis, *argA* and *argC*, while the LSTM-RNN relies on different promoters that are involved in other metabolic or biosynthetic processes.

Biosensor Validation. Given the severe impact of heavy metals on human health (42) and the persistence of water quality issues in the United States (43), we sought to deploy the Dynomics platform as a real-time water quality biosensor. To verify that this device was functional on waters of varying ion compositions, we conducted experiments with media made from municipal water samples from San Diego, Seattle, Chicago, Miami-Dade, and New York City with added cadmium. Fig. 5*A* shows the LSTM-RNN classifier predictions for cadmium exposures on each city's water supply. While there is some misclassification of cadmium as zinc, there are few instances of incorrectly predicting the presence of a toxin versus water, even with largely different water compositions between cities. Additionally, the mean absolute SHAP values for each city correlated strongly with those for laboratory Milli-Q water (R^2 = 0.853), indicating that water composition did not affect gene response. *zntA* was the best predictor of cadmium presence across all water compositions (*SI Appendix*, Fig. S13).

The Dynomics device was also exposed to samples collected from the Gold King Mine spill in August 2015. Fig. 5*B* shows the predictions of the LSTM-RNN classifier on samples from the spill, collected from the San Juan River. The classifier predictions are output as multiclass, multilabel probability vectors. As the sample was introduced onto the device, the probability of uncontaminated water decreased significantly while the probabilities of the other metals increased. The metal with the highest probability, iron, was also the most abundant metal in the samples, as measured by inductively coupled plasma mass spectrometry (ICP-MS) (*SI Appendix*, Table S2). Despite the classifier not being trained on combinations of metals or at the concentrations present in these samples, the ability to reliably report the presence of the most prominent metal and, to a lesser degree, the less abundant metals suggests the broad applicability of this platform for heavy metal detection.

Discussion

In this work, we developed a high-throughput microfluidic platform to track the transcriptional dynamics of thousands of *E. coli* genes in parallel. The Dynomics platform offers a useful experimental approach through its high temporal resolution, degree of multiplexing, and precise experimental control. In

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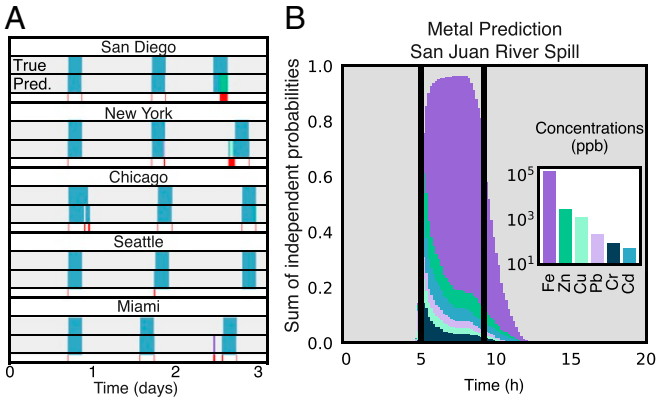

Fig. 5. Dynomics and machine learning on environmental samples. (*A*) LSTM-RNN classification of cadmium contamination in five different urban water sources. Each city has a row for the true media condition, the predicted condition, and whether the time points are misclassified (red). The colors correspond to the metals in *B*, *Inset*. (*B*) Multiclass, multilabel classification of water samples from the San Juan River during the 2015 Gold King Mine waste water spill. Independent probabilities of each class are determined by the sigmoid activation function. The plot shows the sum of the classifier probabilities, averaged across triplicate sample exposures (addition and removal at vertical black lines). *Inset* bar chart shows the concentration of detectable metals in San Juan River samples as determined by ICP-MS. The colors of predicted toxins correspond to the metals plotted in *Inset*.

a high-throughput screen using Dynomics, we simultaneously exposed 1,807 strains of the promoter-based *E. coli* GFP library to nine different heavy metals. The fine-grained temporal gene expression data it produced highlighted the unique dynamics of stimuli-specific genes previously reported as heavy metal responsive (44) and identified gene clusters that shared similar response dynamics.

We illustrate our platform's potential for exploring the dynamics of transcriptional networks by applying machine-learning techniques to examine heavy metal stress responses in *E. coli*. Here we demonstrate that supervised machine learning can infer exposure to environmental stressors from real-time observation of transcriptional activity at the genome scale. Time series from 1,807 strains were used to differentiate between multiple biotic and xenobiotic heavy metals. We believe this study is an informative instance of dynamic mapping between transcriptomic changes captured in live microorganisms on the one hand and their surrounding environment on the other. These data, with genome-scale coverage and high sampling frequency, could be used in future studies to screen large strain libraries for common motifs, such as nonlinear interaction patterns and feedback loops, which are difficult to discern using static gene expression data (7). Furthermore, we use explainable AI techniques to gain insight into the features used by the predictive algorithms trained on our transcriptional data. The SHAP-XAI revealed that formally different algorithms rely on different biological features to classify transcriptomic adaptation to stress. While a decision tree-based model relied heavily on a small number of strains, a better-performing deep-learning algorithm based its prediction on many strains of moderate influence (Fig. 4). These findings reveal that there are different ways to segregate the high-dimensional space explored by an organism's transcriptome during sensory response.

Finally, we show the real-world applicability of our platform for the detection of heavy metals in both urban water sources and field samples from a recent environmental catastrophe. Compared to conventional methods of metal quantification, such as atomic absorption spectroscopy or ICP-MS, the Dynomics platform sacrifices detection sensitivity for the ability to report continuous measurements, eliminating the need to take discrete samples. Although the Dynomics platform sometimes experiences a slight lag in the detection of metals when they are first introduced or removed, the platform is still a significant improvement over grab sampling. While previous approaches to microorganism-based heavy metal sensing have relied on engineering a small number of biosensors that are specific to one metal (45), here we use *E. coli*'s transcriptomic response at the genome scale to detect environmental stressors. Our biosensor was robust to the differences in ionic composition of five urban water sources and consistently detected cadmium in those samples. In addition, it was able to simultaneously detect multiple target metals in mine spill samples, despite not being trained to perform this type of multiclass, multilabel classification. This result suggests our approach may outperform single-purpose biosensors in accuracy and robustness and may be adaptable to more varied sensing tasks via optimization through testing combinations of metals and different concentrations of metals. In summary, combining high-throughput microfluidics and machine learning can produce insights into the coordination of cellular processes at a system level and this type of data can be leveraged for environmental monitoring.

Materials and Methods

Microfluidic Device Development and Fabrication. Our group has previously described the microfabrication techniques used to pattern SU-8 photoresist onto a silicon wafer to create the mold for our device (46). A polydimethylsiloxane (PDMS) device was made from the wafer by mixing 77 g of Sylgard 184 and pouring it on the wafer centered on a level $5'' \times 5''$ glass plate surrounded with an aluminum foil seal. The degassed wafer and PDMS were cured on a flat surface for 1 h at 95 °C.

Cell Preparation. The *E. coli* promoter library (22) was arrayed using the Singer ROTOR Stinger (Singer Instrument Co. Ltd.) attachment from 96-well density formatted agar plates onto four 1,536-density formatted agar plates to match the layout of the cell traps on the microfluidic device. At the time of experimental setup, the four 1,536-density agar plates were combined onto one 6,144-density agar plate using the Singer ROTOR and grown for 2 h at 37 °C before being transferred to the device.

Microfluidic Device Loading and Bonding. A PDMS device cleaned with 70% ethanol and adhesive tape was aligned to a custom fixture compatible with the Singer ROTOR. Both the fixture and a clean $4'' \times 3''$ glass slide sonicated with 2% Hellmanex III were exposed to oxygen plasma. Cells were spotted from the previously arrayed 6,144-density agar plate to the aligned PDMS device using the Singer ROTOR spotting robot. The device and glass slide were bonded together and cured at 37 °C for 2 h.

Experimental Protocol. Microfluidic experiments were performed on a custom optical assembly described in *SI Appendix*. Continuous imaging occurred every 10 min, imaging both the transmitted light and GFP fluorescence channels. Cells were grown in the device on LB media with kanamycin, 0.075% Tween-20, and 50 mM methyl α-D-mannopyranoside until traps were filled to confluence. The media were then switched to a heavy-metal–trace-free minimal media (HM9) minimal media described in *SI Appendix*, Table S1, which was based on a previous study (47) and optimized for microfluidic *E. coli* growth with minimal traces of metals. Cells were grown on HM9 for 48 h before inducing with heavy metals. Heavy metal inductions occurred once a day for 4 h with HM9 media flowing on chip for the remaining 20 h. Quintuplicate inductions of each metal were performed in a random order across multiple experiments, with each experiment lasting 7 to 14 d. A total of 2,176 time traces were collected from each experiment (Fig. 2 *A*–*C*). Extracted time traces were normalized to remove device background fluorescence and strain background fluorescence (*SI Appendix*). Detailed methods on experimental setup and data collection can be found in *SI Appendix*, Table S3.

Municipal Water Experimental Setup. Water samples were obtained from the Department of Water Management at the City of Chicago in Chicago, IL; the Alex Orr Water Treatment Plant in Miami, FL; the New York City Department of Environmental Protection and Bureau of Water Supply in Corona, NY; the Seattle Public Utilities Water Quality Laboratory in Seattle, WA; and

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the Alvarado Water Treatment Plant in San Diego, CA. HM9 media for each city water experiment were prepared by diluting 5× HM9 concentrate made from Milli-Q water with the water obtained from each city. The microfluidic device was initially grown on LB media with kanamycin, 0.075% Tween-20, and 50 mM methyl α-D-mannopyranoside until traps were filled to confluence and then switched to HM9 made with city water for the remainder of the experiment. Cadmium diluted in the HM9 city water media was used to perform inductions as described in *SI Appendix*.

Gold King Mine Spill Experimental Setup. Water was collected from Mexican Hat, UT in August 2015 when the Gold King Mine spill plume reached the collection point in the San Juan River. Samples were stored in 0.5% HCl acid until tested. HM9 media were prepared by diluting 5× HM9 concentrate made from Milli-Q water with filtered San Juan River samples. The pH was adjusted to 7.05. The metal concentrations of the HM9 San Juan River samples were tested by ICP-MS at the Environmental and Complex Analysis Laboratory (ECAL) at University of California, San Diego. Four-hour inductions were performed as described in *SI Appendix*.

Machine-Learning and Data Analysis Methods. We transformed our 18 standardized experiments' time points into a first derivative-based feature for the training and testing feature sets. To optimize the classifiers, extensive Bayesian optimization searches were used to find optimal hyperparameter combinations (48). Throughout our hyperparameter searches, we used leave-one-out cross-validation on a per-experiment basis and appropriate overfitting-prevention strategies to ensure that any resultant classifier would generalize to future datasets. All classifiers were evaluated using the F_1-macro scoring metric. The F_1-macro score, which is the per-class average of the harmonic mean of precision and recall, was especially well suited because of our dataset's large multiclass imbalances, with water making up ∼86% of the final feature set (49). Finally, all generalization evaluations were performed by recording the results of using leave-one-out cross-validation with early stopping and then taking the mean prediction across the cross-validation's output.

Data Availability. Preprocessed, labeled machine-learning features, the corresponding library strain position records, and the relevant metadata and code for our experiments are available on the University of California San Diego Biodynamics Laboratory website (http://biodynamics.ucsd.edu/downloads).

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1. B. Kholodenko, M. B. Yaffe, W. Kolch, Computational approaches for analyzing information flow in biological networks. *Sci. Signal.* **5**, re1 (2012).
2. R. Milo *et al.*, Network motifs: Simple building blocks of complex networks. *Science* **298**, 824–827 (2002).
3. F. Jacob, J. Monod, Genetic regulatory mechanisms in the synthesis of proteins. *J. Mol. Biol.* **3**, 318–356 (1961).
4. T. S. Gardner, C. R. Cantor, J. J. Collins, Construction of a genetic toggle switch in Escherichia coli. *Nature* **403**, 339–342 (2000).
5. M. Krupp *et al.*, RNA-Seq Atlas-a reference database for gene expression profiling in normal tissue by next-generation sequencing. *Bioinformatics* **28**, 1184–1185 (2012).
6. G. La Manno *et al.*, RNA velocity of single cells. *Nature* **560**, 494–498 (2018).
7. D. L. Shis, M. R. Bennett, O. A. Igoshin, Dynamics of bacterial gene regulatory networks. *Annu. Rev. Biophys.* **47**, 447–467 (2018).
8. N. T. Ingolia, S. Ghaemmaghami, J. R. S. Newman, J. S. Weissman, Genome-wide analysis in vivo of translation with nucleotide resolution using ribosome profiling. *Science* **324**, 218–223 (2009).
9. Y. Ho *et al.*, Systematic identification of protein complexes in Saccharomyces cerevisiae by mass spectrometry. *Nature* **415**, 180–183 (2002).
10. D. A. Lashkari *et al.*, Yeast microarrays for genome wide parallel genetic and gene expression analysis. *Proc. Natl. Acad. Sci. U.S.A.* **24**, 13057–13062 (1997).
11. M. J. Heller, DNA microarray technology: Devices, systems, and applications. *Annu. Rev. Biomed. Eng.* **4**, 129–153 (2002).
12. N. Hao, B. A. Budnik, J. Gunawardena, E. K. O'Shea, Tunable signal processing through modular control of transcription factor translocation. *Science* **339**, 460–464 (2013).
13. J. E. Purvis, G. Lahav, Encoding and decoding cellular information through signaling dynamics. *Cell* **152**, 945–956 (2013).
14. M. R. Bennett *et al.*, Metabolic gene regulation in a dynamically changing environment. *Nature* **454**, 1119–1122 (2008).
15. J. Uhlendorf *et al.*, Long-term model predictive control of gene expression at the population and single-cell levels. *Proc. Natl. Acad. Sci. U.S.A.* **109**, 14271–14276 (2012).
16. J. T. Mettetal, D. Muzzey, C. Gomez-Uribe, A. van Oudenaarden, The frequency dependence of osmo-adaptation in Saccharomyces cerevisiae. *Science* **319**, 482–484 (2008).
17. N. Dénervaud *et al.*, A chemostat array enables the spatio-temporal analysis of the yeast proteome. *Proc. Natl. Acad. Sci. U.S.A.* **110**, 15842–15847 (2013).
18. R. Zhang *et al.*, High-throughput single-cell analysis for the proteomic dynamics study of the yeast osmotic stress response. *Sci. Rep.* **7**, 42200 (2017).
19. Y. Taniguchi *et al.*, Quantifying E. coli proteome and transcriptome with single-molecule sensitivity in single cells. *Science* **329**, 533–539 (2010).
20. A. Prindle *et al.*, A sensing array of radically coupled genetic 'biopixels'. *Nature* **481**, 39–44 (2012).
21. C. Zhang *et al.*, Ultra-multiplexed analysis of single-cell dynamics reveals logic rules in differentiation. *Sci. Adv.* **5**, eaav7959 (2019).
22. A. Zaslaver *et al.*, A comprehensive library of fluorescent transcriptional reporters for Escherichia coli. *Nat. Methods* **3**, 623–628 (2006).
23. R. Sharma, C. Rensing, P. Rosen, B. Mitra, B. P. Rosen, The ATP hydrolytic activity of purified ZntA , a Pb(II)/Cd(II)/Zn(II)-translocating ATPase from Escherichia coli. *J. Biol. Chem.* **275**, 3873–3878 (2000).
24. G. Grass, C. Rensing, CueO is a multi-copper oxidase that confers copper tolerance in Escherichia coli. *Biochem. Biophys. Res. Commun.* **286**, 902–908 (2001).
25. S. Ghatak, Z. A. King, A. Sastry, B. O. Palsson, The y-ome defines the 35% of Escherichia coli genes that lack experimental evidence of function. *Nucleic Acids Res.* **47**, 2446–2454 (2019).
26. G. Graham, N. Csicsery, E. Stasiowski, G. Thouvenin, Labeled data set for "Genome-scale transcriptional dynamics and environmental biosensing." http://biodynamics.ucsd.edu/downloads. Deposited 11 December 2019.
27. T. Chen, C. Guestrin, XGBoost: A scalable tree boosting system. ArXiv:1603.02754 (10 June 2016).
28. S. Hochreiter, J. Schmidhuber, Long short-term memory. *Neural Comput.* **9**, 1735–1780 (1997).
29. J. P. McHugh *et al.*, Global iron-dependent gene regulation in Escherichia coli. *J. Biol. Chem.* **278**, 29478–29486 (2003).
30. C. Rensing, B. Mitra, B. P. Rosen, The zntA gene of Escherichia coli encodes a Zn(II)-translocating P-type ATPase. *Biochemistry* **94**, 14326–14331 (1997).
31. S. P. Singh *et al.*, Machine learning based classification of cells into chronological stages using single-cell transcriptomics. *Sci. Rep.* **8**, 17156 (2018).
32. D. Castelvecchi, Can we open the black box of AI? *Nat. News* **538**, 20–23 (2016).
33. J. Ma *et al.*, Using deep learning to model the hierarchical structure and function of a cell. *Nat. Methods* **15**, 290–298 (2018).
34. J. H. Yang *et al.*, A white-box machine learning approach for revealing antibiotic mechanisms of action. *Cell* **177**, 1649–1661.e9 (2019).
35. J. Zhou, O. G. Troyanskaya, Predicting effects of noncoding variants with deep learning-based sequence model. *Nat. Methods* **12**, 931–934 (2015).
36. S. Lundberg, S.-I. Lee, A unified approach to interpreting model predictions. ArXiv:1705.07874 (25 November 2017).
37. S. M. Lundberg, G. G. Erion, S.-I. Lee, Consistent individualized feature attribution for tree ensembles. ArXiv:1802.03888 (7 March 2019).
38. S. M. Lundberg *et al.*, Explainable AI for trees: From local explanations to global understanding. ArXiv:1905.04610 (11 May 2019).
39. L. S. Shapley, "A value for n-person games" in *Contributions to the Theory of Games*, H. W. Kuhn, A. W. Tucker, Eds. (Princeton University Press, 1953), vol. 2, pp. 307–317.
40. S. Dodge, L. Karam, A study and comparison of human and deep learning recognition performance under visual distortions. https://ieeexplore.ieee.org/abstract/document/8038465. Accessed 25 May 2019.
41. A. C. Esteves, J. Felcman, Study of the effect of the administration of Cd(II) cysteine, methionine, and Cd(II) together with cysteine or methionine on the conversion of xanthine dehydrogenase into xanthine oxidase. *Biol. Trace Elem. Res.* **76**, 19–30 (2000).
42. P. B. Tchounwou, C. G. Yedjou, A. K. Patlolla, D. J. Sutton, "Heavy metal toxicity and the environment" in *Molecular, Clinical and Environmental Toxicology*, A. Luch, Ed. (Springer, Basel, 2012), pp. 133–164.
43. M. Allaire, H. Wu, U. Lall, National trends in drinking water quality violations. *Proc. Natl. Acad. Sci. U.S.A.* **115**, 2078–2083 (2018).
44. S. P. LaVoie, A. O. Summers, Transcriptional responses of Escherichia coli during recovery from inorganic or organic mercury exposure. *BMC Genom.* **19**, 52 (2018).
45. H. J. Kim, H. Jeong, S. J. Lee, Synthetic biology for microbial heavy metal biosensors. *Anal. Bioanal. Chem.* **410**, 1191–1203 (2018).
46. M. S. Ferry, I. A. Razinkov, J. Hasty, Microfluidics for synthetic biology: From design to execution. *Methods Enzymol* **497**, 295–372 (2011).
47. R. A. LaRossa, D. R. Smulski, T. K. Van Dyk, Interaction of lead nitrate and cadmium chloride with Escherichia coli K-12 and Salmonella typhimurium global regulatory mutants. *J. Ind. Microbiol.* **14**, 252–258 (1995).
48. B. Shahriari, K. Swersky, Z. Wang, R. P. Adams, N. De Freitas, Taking the human out of the loop: A review of Bayesian optimization. *Proc. IEEE* **104**, 148–175 (2016).
49. Z. C. Lipton, C. Elkan, B. Narayanaswamy, Thresholding classifiers to maximize F1 score. ArXiv:1402.1892 (14 May 2014).

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Supporting Information Text

SI Materials and methods

Microfluidic device design. All microfluidic devices were designed on AutoDesk AutoCAD for Mac and Windows.

Wafer Fabrication. The silicone wafer was fabricated using standard photolithography techniques previously described by our group (1). The cell trap layer was fabricated using 2005 SU-8 photoresist with a spin speed of 5000 rpm and had a resulting height of 3.85-4.05 μms, the minor channel layer was fabricated using 2075 SU-8 photoresist with a spin speed of 4250 rpm and had a resulting height of 45-50 μms, and the major channel layer was fabricated using 2075 SU-8 photoresist with a spin speed of 1200 on top of an undeveloped major channel layer, resulting in a final height of 200-260 μms.

Microfluidic device fabrication. A poly-dimethylsiloxane (PDMS) device was made from the wafer by mixing 77 grams of Sylgard 184 and pouring it on the wafer centered on a level 5"× 5" glass plate surrounded with an aluminum foil seal. The degassed wafer and PDMS was cured on a flat surface for one hour at 95°C.

Cell preparation. The broad experimental workflow, shown in Figure S1, begins with the maintenance of the *E. coli* library to be spotted on the microfluidic devices. *E. coli* used in these experiments were initially stored at -80°C in a LB + 20% glycerol solution on 384-well microtiter plates. Plates were thawed for 2 hours and spotted at a 384 SBS density to LB+kanamycin agar plates using a Singer ROTOR. After overnight growth at 37°C, cells were arrayed using the Singer ROTOR Stinger attachment from 384-well density formatted LB+kanamycin agar plates onto four 1536-density formatted LB+kanamycin agar plates to match the layout of the cell traps on the microfluidic device. After overnight growth at 37°C, these plates were stored at 4°C for up to 2 weeks.

To load microfluidic devices, four stored 1536 SBS-density agar plates were re-spotted and combined onto a single 6144 SBS-density agar plate with up to 2,176 cells spots. These cells were allowed to grow for 2 hours at 37°C before being transferred to a microfluidic device.

Microfluidic device loading and bonding. A PDMS device cleaned with 70% Ethanol and adhesive tape was aligned to a custom fixture compatible with the Singer ROTOR. Both the fixture and a clean 4"× 3" glass slide sonicated with 2% Helmanex III were exposed to oxygen plasma. Cells were spotted from the previously arrayed 6144-density agar plate to the aligned PDMS device using the Singer ROTOR spotting robot. The device and glass slide were bonded together and cured at 37°C for two hours.

Experimental set-up. Before setting up a microfluidic experiment, the device was placed in a vacuum for a minimum of 20 minutes. The device was then mounted onto the custom optical enclosure. The inlet port was connected to a 140 mL syringe (Covidien Monoject Syringe) and PTFE tubing (Cole Palmer PTFE#24 AWG tubing) with LB media with kanamycin, 0.075% Tween-20, and 50 mM methyl α-D-mannopyranoside. The waste port was connected to PTFE tubing and a 1L waste bottle. The height difference between the inlet and outlet was 39" corresponding to a flow rate of 11 mL/hr. Tween-20 and methyl α-D-mannopyranoside were used in the media to prevent biofilming and thereby increase the longevity of microfluidic experiments. Tween-20 has been used by our group in many experiments without an adverse effect on *E. coli* (1, 2). Methyl α-D-mannopyranoside inhibits normal surface attachment of type I pili but cannot be metabolized by *E. coli* (3). After 16-24 hours of growth on LB media for cells to fill the traps, the media was switched to an HM9 minimal media described in Table S1, which was based on a previous study (4) and optimized for microfluidic *E. coli* growth with minimal traces of metals. Cells were grown on HM9 for 48 hours before inducing with heavy metals.

Induction protocol. Heavy metal inductions occurred once a day for four hours. To induce, the HM9 in the inlet syringe was slowly pipetted out of the syringe and the HM9 + metal media was slowly pipetted in. To remove the heavy metal media after the 4 hour induction, the HM9 + metal media was pipetted out, and the remaining dead volume was washed with 2×5 mL HM9 wash steps. Then the HM9 media was pipetted in to fill the syringe. The order of daily metal inductions were randomized for each experiment. The number of inductions and concentrations for each metal is listed in Table S4. To validate our induction protocol across the full device, a Dynomics chip was spotted with 2,176 replicates of the *zntA* promoter strain and subjected to multiple replicate cadmium inductions (Fig. S4). Strains across the device maintained statistically similar response parameters (amplitude of response and relaxation time) regardless of day of induction or spatial positioning on the device.

Municipal water experimental set-up. Water samples were obtained from the Department of Water Management at the City of Chicago, in Chicago, IL, the Alex Orr Water Treatment Plant in Miami, FL, the New York City Department of Environmental Protection and Bureau of Water Supply in Corona, NY, the Seattle Public Utilities Water Quality Lab in Seattle WA, and the Alvarado Water Treatment Plant in San Diego, CA. HM9 media for each city water experiment was prepared by diluting 5× HM9 concentrate made from Milli-Q water with the water obtained from each city. The microfluidic device was initially grown on LB media with kanamycin, 0.075% Tween-20, and 50 mM methyl α-D-mannopyranoside until traps were filled to confluence and then switched to HM9 made with city water for the remainder of the experiment. Cadmium diluted in the HM9 city water media was used to perform inductions.

Gold King Mine spill experimental set-up. Water was collected from the San Juan River (SJR) where it flowed through Mexican Hat, Utah on August 12-13, 2015. Sampling began at 12:30 PM on Aug. 12 and was taken in triplicate every 6 hours from a well-mixed region of the main flow, just upstream of the Mexican Hat Bridge. It was later confirmed by the Utah Department of Environmental Quality that this period was when the Gold King Mine Spill plume reached the collection point in the SJR. Samples were taken in pre-acid-washed amber, glass sampling bottles and immediately spiked with 50 mL of 10% HCl acid. All samples were kept on ice until tested. HM9 media was prepared by diluting 5× HM9 concentrate made from Milli-Q water with filtered SJR samples. The pH was adjusted to 7.05. The metal concentrations of the SJR and HM9 samples were tested by ICP-MS at the Environmental and Complex Analysis Laboratory (ECAL) at UC San Diego. Four hour inductions were performed as described above.

Live-cell imaging and data. Microfluidic devices were imaged in a custom optical enclosure continuously every ten minutes in both the transmitted light and GFP fluorescence channels with a 1 second and 60 second exposure, respectively. The custom optical enclosure uses an SBIG STX-16803 CCD Camera with a custom lens stack assembly containing the Semrock FF01-466/40-32-D-EB and FF02-520/28-50-D-EB excitation and emission filters, respectively. The enclosure has green and blue LED spotlight sources obtained from ProPhotonix for transmitted light and fluorescence light sources, respectively. The optical resolution of the enclosure is 36 μm. The enclosure was temperature controlled to 37°C.

Images were synced from the enclosure to a server via WiFi for further data processing. Custom software produced flat-field-corrected images in both channels in real-time to remove optical vignetting using the following equation:

$$C = m * \frac{R - D}{F' - D'} \quad [1]$$

where R is the raw image to be flat-field corrected (Fig. S2A), D is the dark-current image for that device, taken at the same exposure settings as R, F' is a raw image taken by the camera with no device present, D' is the dark-current image taken at same exposure as F', m is the mean value for all values in the array $(F' - D')$, and C is the resulting corrected image (Fig. S2B).

Flat-field corrected images were then registered to an extraction mask to account for the device drifting from thermal expansion, with mean transmitted light and green fluorescence channels from masked bulb and background regions extracted (Fig. S3).

Data was initially processed by subtracting the local background signal, in order to eliminate any local or regional fluctuations that are of an additive (or, analogously, subtractive) nature. Then, by dividing by the same background signal, we form a measure of the amplification of the signal over the local background. The result of this background correction was to produce a vector $\vec{x_t}$ representing the background-corrected fluorescent signals of all device positions at time t:

$$x_{(t,\ s_i)} = \frac{x^{bulb}_{(t,\ s_i)} - x^{background}_{(t,\ s_i)}}{x^{background}_{(t,\ s_i)}} \quad [2]$$

where t refers to the current time point, s_i refers to the strain in device position i, $x^{bulb_{(t,\ s_i)}}$ is the flat-field corrected fluorescent signal from the *bulb* region of position i at time t, and $x^{background}_{(t,\ s_i)}$ refers to the flat-field corrected local background fluorescent signal at position i at time t.

For fold change analysis, fluorescence values were passed through a median filter (scipy.signal.medfilt, kernel_size=11) and normalized by promoterless strains (Fig. S5). Promoterless strains U139 and U66 from the GFP promoter library (5) were spotted at various locations across each device, with 20 device positions of each, for 40 in total. Normalized fluorescence for Figure 2 was thus determined as:

$$x^{normalized}_{(t,\ s_i)} = \frac{scipy.signal.medfilt(x_{(t,\ s_i)}) - mean(scipy.signal.medfilt(x^{promoterless}_{(t,\ s_i)}))}{mean(scipy.signal.medfilt(x^{promoterless}_{(t,\ s_i)}))} \quad [3]$$

Fold change in Figure 2D was calculated as the quotient of the normalized fluorescence at the first and last time point of each metal exposure. P-values were determined by a dependent two-sided t-test on the the log_2 fluorescent values at the start and end of each metal exposure (scipy.stats.ttest_rel).

Gene ontology results for Figure 2. We performed enrichment analysis (6) on gene clusters obtained via agglomerative clustering of promoter activity, normalized between 0 and 1, of 1,995 strains responding to a 50 ppb Zn induction (see main text Fig.2A). Figure S14 represent the enriched GO term obtained for each cluster via http://pantherdb.org/. The statistical method used was Fisher's exact test with no correction for multiple testing. No correction was selected as we chose to minimize the number of false negatives over the number of false positives. We note that this increases the likelihood of finding enrichment terms by chance only. The reference list for the GO enrichment analysis was the 1,807 unique genes of the Zaslaver library (5).

Engineering of machine learning features. All data used for machine learning results were pre-processed into first derivative-based features. Intuitively, a first derivative-based feature is an excellent candidate for any sort of machine learning model because it is what the human brain instinctively monitors when looking for changes in strain-promoter behavior (7). Any significant modification in the mean or variance of the first derivative of a given promoter while induced or uninduced could

signify that the promoter is sensitive to that particular environmental perturbation. The engineering of a first derivative-approximation feature essentially amounts to distilling out the pure changes in the original feature's behavior, while effectively filtering out any changes that could be due to extraneous local or global environmental influences. Figure S6 illustrates the results of the feature engineering process.

Raw data was converted into first-derivative features using the following series of transformations:

Background signal removal. Background signal removal was performed as described by Eqn. 2. The result of this background correction was to produce a vector $\vec{x_t}$ representing the background-corrected fluorescent signals of all device positions at time t:

$$\begin{bmatrix} x_{(t,\ s_0)} \\ x_{(t,\ s_1)} \\ \vdots \\ x_{(t,\ s_{2174})} \\ x_{(t,\ s_{2175})} \end{bmatrix} = \vec{x_t} \quad [4]$$

Z-scoring by time point. In order to account for slight differences in mean fluorescence across multiple optical enclosures, we used the following per time point z-score:

$$\frac{\vec{x}_t - \overline{x}_t}{\sigma_{\vec{x}_t}} = \vec{x}'_t \quad [5]$$

where $\vec{x}_t$ is the vector of the background-corrected fluorescent signal for all occupied strain positions at time t; $\overline{x}_t$ is the mean value for the vector at the same time point; and $\sigma_{\vec{x}_t}$ is the standard deviation of $\vec{x}_t$.

Exponential smoothing. Since we ultimately wanted to use a first derivative-based feature, we first smooth the signal prior to calculating the first finite-difference. The smoothing prevents the amplification of noise in the differentiation step. To smooth, we implemented the Holt-Winters second-order exponentially weighted moving average (HWMA), which de-noises the data based on the history of both the trajectory and the history of the trajectory's derivative. We chose to use the HWMA over other smoothing methods because it is rear-facing only, which means that it smooths the current time point based only on the previous time points. This is as opposed to other methods, which require future time points to adequately smooth the current time point. HWMA, therefore, is uniquely suited to real-time sensing applications in which dramatic changes in the derivative of the trajectory can occur (8).

$$\begin{aligned} \vec{x}''_1 &= \vec{x}'_1 \\ \dot{\vec{x}}''_1 &= \vec{x}'_1 - \vec{x}'_0 \\ &\vdots \\ \vec{x}''_t &= \alpha\vec{x}'_t + (1-\alpha)(\vec{x}''_{t-1} + \dot{\vec{x}}''_{t-1}) \\ \dot{\vec{x}}''_t &= \beta(\vec{x}''_t - \vec{x}''_{t-1}) + (1-\beta)\dot{\vec{x}}''_{t-1} \end{aligned} \quad [6]$$

where $\vec{x}''_t$ is the smoothed version of $\vec{x}'_t$, and $\dot{\vec{x}}''_t$ is that point's first-finite difference. Given that samples were taken at a consistent frequency, $\dot{\vec{x}}''_t$ then becomes an approximation for the derivative of the trajectory at time t. α and β are weights that were empirically chosen from the interval $[0, 1]$ and were constant for all experiments. α can be thought of as the relative importance placed on the most recent values, as opposed to values from further in the past. Similarly, β is the relative importance placed upon the function's most recent approximation of the first derivative.

First Finite Difference. The penultimate feature-engineering step was to take the first finite difference:

$$\dot{\vec{x}}_t = \vec{x}'_t - \vec{x}'_{t-1} \quad [7]$$

Vector normalization. The final step that was used in creating our finished features for machine learning was to normalize each time point vector by itself.

$$\left(\frac{2176}{\sum_{i=0}^{2175} \dot{x}_{(t,\ s_i)}} \right) \cdot \dot{\vec{x}}_t = \dot{\vec{X}}_t \quad [8]$$

Since most derivatives had the characteristic that, for the majority of time points t, $\dot{x}_{(t,\ s_i)} << 1$, normalizing each time point by the mean of itself had the effect of making the new, normalized vector's mean equal to 1. First derivative values that were close to 0 remained close to 0, while the magnitude of larger values was amplified.

Additional feature preprocessing. In addition to the data preprocessing and feature engineering that were explicitly enumerated in the previous subsection, other steps were often taken when training and testing any machine learning algorithms. These additional steps each dropped some portion of the final feature set, but *only after* the features had been calculated using the entire original experiments. Dropping these time points before calculating the features would have introduced potential discontinuities to any features approximating the first-derivative. All features were processed and cached in permanent memory. Only the cached features were used for any further analysis.

Trimming grow-up phase and end phase time points from feature sets. All experiments included transient periods over the first several days of the experiment. These transients were caused by the colonies' recoveries post-spotting shock, their growth-to-effluence within their individual traps, and their second recovering following the switch to minimal HM9 growth media. In addition, since most experiments were run until the microfluidic chips were deemed unusable due to clogging by biofilms, the final hours of most experiments did not yield high-quality water data. Since the beginnings and the ends of the experiments represented non-steady state local and global growth conditions, the features from these periods were dropped prior to analysis and machine learning. This trimming was done by removing features more than eighteen hours before the first induction and more than four hours after the final metal induction.

Dropping bad, unwanted, and special inductions. In addition, inductions using undetectable metals, such as arsenic, mercury, and antimony, and special inductions that introduced any non-standard inducers were dropped from the feature set post-processing.

Machine learning.

Bayesian optimization of hyperparameters. In order to optimize both the XGBoost and LSTM-RNN classifiers, extensive Bayesian Optimization searches were implemented to find optimal hyperparameter combinations (9). The popular Bayesian optimization library *hyperopt* was used to conduct the searches (10). A general search harness was coded and searches were conducted using randomized leave-three-out cross-validation over a total of 1000 hyperparameter candidate sets.

Cross-validated classifier evaluation with multiclass F_1 metric. Once a suite of optimal hyperparameters had been identified, we used leave-one-out cross-validation on a per-experiment basis to estimate classifier generalization to unseen data sets. Early stopping was used for both classification algorithms to prevent overfitting. To evaluate early stopping, each experiment from the training set served as the early stopping test set. Once the early stopping termination criteria had been met, the resultant classifier would be evaluated on the validation experiment; this process was repeated for each experiment in the training set and the test results were averaged together. All classifiers were evaluated using the F_1–macro scoring metric. The F_1–macro score, which is the per-class average of the harmonic mean of precision and recall, was especially well-suited because of our data set's large multiclass imbalances, with water making up approximately 86% of the final feature set (11) (Fig. S7).

XGBoost training. Extreme gradient boosted trees (XGBoost) are a tree ensemble method invented by Tianqi Chen in 2014. As opposed to random forests, which build their ensembles using bootstrap aggregating and additional randomization strategies, XGBoost builds its tree ensembles in an additive manner. We chose to use XGBoost because of its reputation for dealing proficiently with high-dimensional, nonlinear data sets such as ours.

All XGBoost functions were implemented in Python. All hyperparameter searches and cross-validation rounds were parallelized and run on a 24-core 2012 Mac Pro with 48 GB RAM. Hyperparameter optimization took place both by hand and with a Bayesian optimizer. Bayesian optimization proceeded as previously described.

The hand-tuning of the XGBoost classifiers, which yielded competent learners, is enumerated below. At each step, we either exhaustively rotated through each experiment as an early stopping validation set to reduce over-fitting or we randomly shuffled and split experiments into train-validate-test groups, depending on the computation times required. Upon the stopping criteria being met, we tested the trained classifier using leave-one-out cross-validation. After rotating through all train-validate-test combinations of experiments, we averaged the results to produce the final performance score for a specific parameter set.

1. We optimized the number of trees for a learning rate of 0.1;
2. Then, we used a grid search to optimize the maximum tree depth and the minimum child weight over reasonable ranges of both parameters. If any values at an extreme limit of a range were selected, the grid-search was repeated over a appropriately-shifted ranges of both parameters;
3. Grid search was then used to find the optimal value for the regularization parameter γ;
4. The number of trees in the ensemble was then re-tuned with early-stopping;
5. We performed a grid search to optimize the subsample and column-sample-by-tree parameters;
6. The final classifier parameter set was then cross-validated over the entire experiment set to produce the final evaluation metric score and multiclass confusion matrix.

In machine learning, the hand-tuning of hyperparameters is typically the realm of experts and is specific to the type of learning algorithm and intrinsic characteristics of the data set in question. It should be noted that such manual hyperparameter tuning can be useful when the researcher is seeking to rapidly explore a classifier space. However, it is usually produces inferior results to other search methods, such as random search and Bayesian optimization. We used the above hand-tuning method to create an estimate of baseline XGBoost classifier performance.

LSTM-RNN training. To build our deep learning network, we used the Python APIs of the popular libraries *tensorflow* and *keras* (12, 13). All training of LSTM-RNN learners, including the Bayesian hyperparameter optimization search, was performed on a cloud-based Paperspace virtual Linux Ubuntu server with 8 CPUs, 30 GB RAM, 250 GB of storage space, and an NVIDIA Tesla V100 GPU, which itself had 16 GB RAM. All backpropagation steps were executed on the GPU. The result was a stateless LSTM-RNN (ie, the LSTM-RNN would reset the cell state to 0 between inputs) with a 80-node LSTM-RNN layer, followed by a 763-node feed-forward layer, and finally a 7-node output layer utilizing a softmax activation function. Regularization strategies were utilized during training to avoid overfitting.

In order to avoid additional computational and, more important, memory load during training, all features were generated and cached on-disc before training began. Since the LSTM-RNN was stateless (ie, it reset its cell state between discrete inputs), the input features were $(num_timesteps) \times (num_strain_positions)$ matrices, with an associated class label that was the label of the middle time point (14). Thus, each input can be thought of as time window, where each row represents a single time point of an experiment and each column is the time series of length $num_timesteps$ of a specific strain-position. For this project, the length $num_timesteps$ was always odd; therefore, there was always an equal number of time points before and after the middle point in each window. As a result, each time window could be thought of as representing that middle point and was accordingly labeled with the point's class. Since the number of time points to be used as an input was also a hyperparameter in need of optimization, a range of time windows were generated and cached in permanent memory. We then implemented a range of custom Python generator classes and incorporated into the training harnesses (15). In any training or prediction call, a new instance of the generators would be passed to a model and would load the pre-cached features and labels on-the-fly. At the end of these calls, the data would be dumped from memory, thereby preventing out-of-memory errors.

Classification of standardized experiments. The Bayesian hyperparameter optimization and subsequent training of both LSTM-RNN and XGBoost learners were performed solely on data from the 18 standardized experiments. No data from either the urban water samples nor the San Juan River samples were included in either case. The only metals used in either case were cadmium, lead, copper, chromium, iron, and zinc.

The confusion matrices in Fig. S8 and Fig. S9 represent the cross-validation results of our optimized LSTM-RNN and XGBoost classifiers, respectively. The recall and precision results are dramatically better for the LSTM-RNN classifier as compared to XGBoost. Although recall is not perfect, the LSTM-RNN rarely (at most 1% of the time) mistakes any metal for water when the metal is actually present. This result is an especially desirable one for a sensor: even if it is wrong about which metal is present, it is still detecting that something is contaminating the water. Similarly, the high rate of precision for water, at approximately 98%, is another desirable sensor quality, as it indicates a low false positive rate. However, some metals were predicted more poorly such as lead, for which the classifier incorrectly predicted no toxin 48% of the time lead was present. This is largely due to the weak promoter responses induced by 0.03 ppm lead, which is only double the Environmental Protection Agency (EPA) maximum contaminant level. Fig. S10 depicts cases of both good and poor lead detection. Fluorescence responses and machine learning features in response to lead inductions were much smaller than cadmium, a higher-performing metal. In both good (66.7% percent of Pb(II) time points correctly classified) and poor (12.5% correctly classified) prediction cases, Pb(II) is not predicted at the start of the induction window, though is predicted at some point during the 4 hour exposure. This means that while there may be a longer delay for weaker metals, the classifier is still able to predict the presence of those metals within a four hour window.

LSTM-RNN classification of urban water samples. All predictions on urban water sample experiments were made using a multiclass, single-label LSTM-RNN that had been fit on the eighteen standardized experiments. Its hyperparameters were the optimal set identified by the Bayesian optimization search, with the softmax activation function being retained in the final layer. The softmax was retained as the activation function because it enables multiclass, single-label classification:

$$\sigma(\mathbf{z})_i = \frac{e^{z_i}}{\sum_{j=1}^{K} e^{z_j}}, \; z = (z_1, \ldots, z_K) \in \mathbb{R}^K \qquad [9]$$

In Eqn. 9, K is the number of classes. This function forces all values to nearly equal 0 or to nearly equal 1, while requiring that the entire function sum to unity. These properties force the softmax function to effectively predict a single class at each output. Eighteen rounds of training and predictions were performed using early stopping to prevent overfitting, with each of the eighteen standardized experiments serving once as the early stopping routine's test set. Predictions were then averaged.

LSTM-RNN classification of San Juan River samples. Since the San Juan River water samples represented a multiclass, multilabel classification task, the LSTM-RNN's output layer's activation function was changed from the softmax function to the sigmoid activation function. Prediction tests were then performed in the same manner as for the urban water samples. Unlike the softmax function (Eqn. 9), the sigmoid activation function's outputs are independent of one another (ie, not constrained to sum to unity):

$$\sigma(\mathbf{z}) = \frac{1}{1 + e^{-\mathbf{z}}} \quad [10]$$

Since they are constrained to the interval $(0, 1)$, they make an excellent choice in a multiclass, multilabel classification problem where probabilities of presence can sum to more than 1. When depicting the results in the main paper, we divided each class probability value by the sum of the class values, thereby transforming guaranteeing that all output probabilities were converted to relative probabilities summing to one.

Model-specific SHAP algorithms. We trained our XAI models over both our XGBoost and LSTM-RNN learners, using Lundberg and Lee's Python library *shap* (16). In order to introspect the tree ensemble learners that trained on the standardized experiments, we utilized *shap*'s TreeExplainer. TreeExplainer leverages the structure of the component trees to greatly increase the calculation of the model's SHAP values (17). Analysis of these SHAP values was conducted in Jupyter Notebooks on a 2012 Mac Pro with 24-cores and 48 GB RAM.

Similarly, the SHAP explainer models for the LSTM-RNN deep learners was built using *shap*'s DeepExplainer algorithm (18, 19). All training was conducted on the aforementioned cloud server, in order to utilize the server's GPU. As a result of the algorithm's memory requirements, another custom generator class was created and training was only performed on an experiment-by-experiment basis. Training a separate explanation model on each experiment is not a problem, however, as a consequence of the Shapley values' linearity property. All calculated SHAP values were cached in permanent memory for subsequent analysis.

Results of arsenic and mercury inductions. Mercury, arsenic, and antimony exposures were also fed into the classifiers, but despite extensive feature-engineering efforts, no classifier was able to successfully detect these metals. This negative result was most likely a consequence of the absence of the *E. coli ars* and *mer* operons from the Alon promoter library (20). These operons are known to be highly efficient at sensing and exporting arsenic and mercury, respectively, from their cells (21, 22). In addition, upon further review, we found that the tested concentrations of both metals were over an order of magnitude lower than the known in-batch minimum inhibitory concentrations for *E. coli* (23). To focus classification on detectable metals, features during these metal inductions were dropped from the data set.

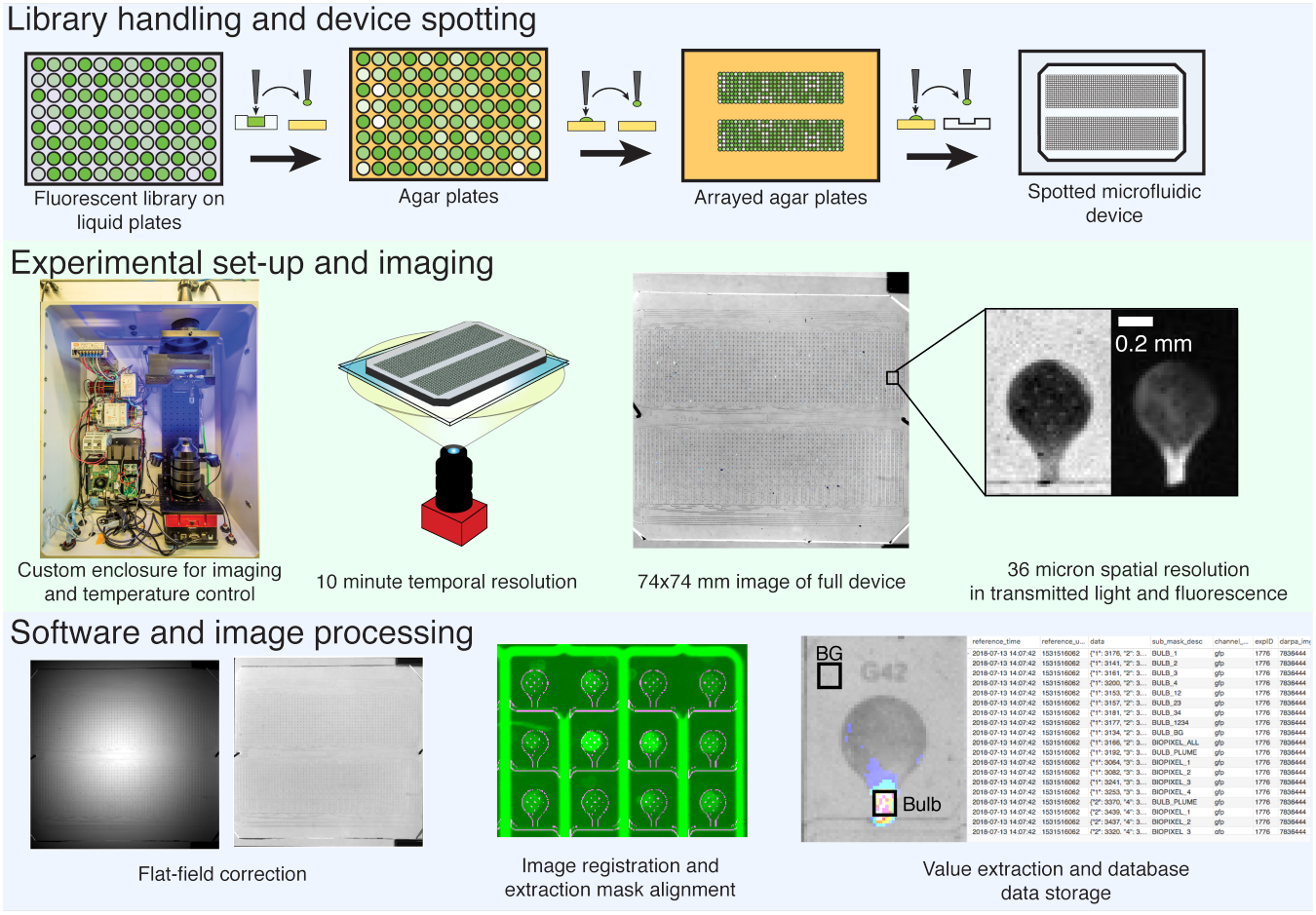

Fig. S1. The workflow of a dynomics experiment. 96-well liquid plates containing *E. coli* library members are respotted and re-arrayed on Agar plates and then transferred onto microfluidic devices with the Singer ROTOR. Spotted chips are placed into a custom optical enclosure, which maintains temperature at 37°C while imaging the full chip once every 10 minutes in both fluorescence and transmitted light channels with a 36 micron resolution. As images are synced to our servers, they are initially flat field corrected. Images are registered to account for xy-drift and an extraction masks is aligned to the device features. Fluorescence and transmitted light values are extracted from extraction regions based on the mask alignment and stored in an SQL database.

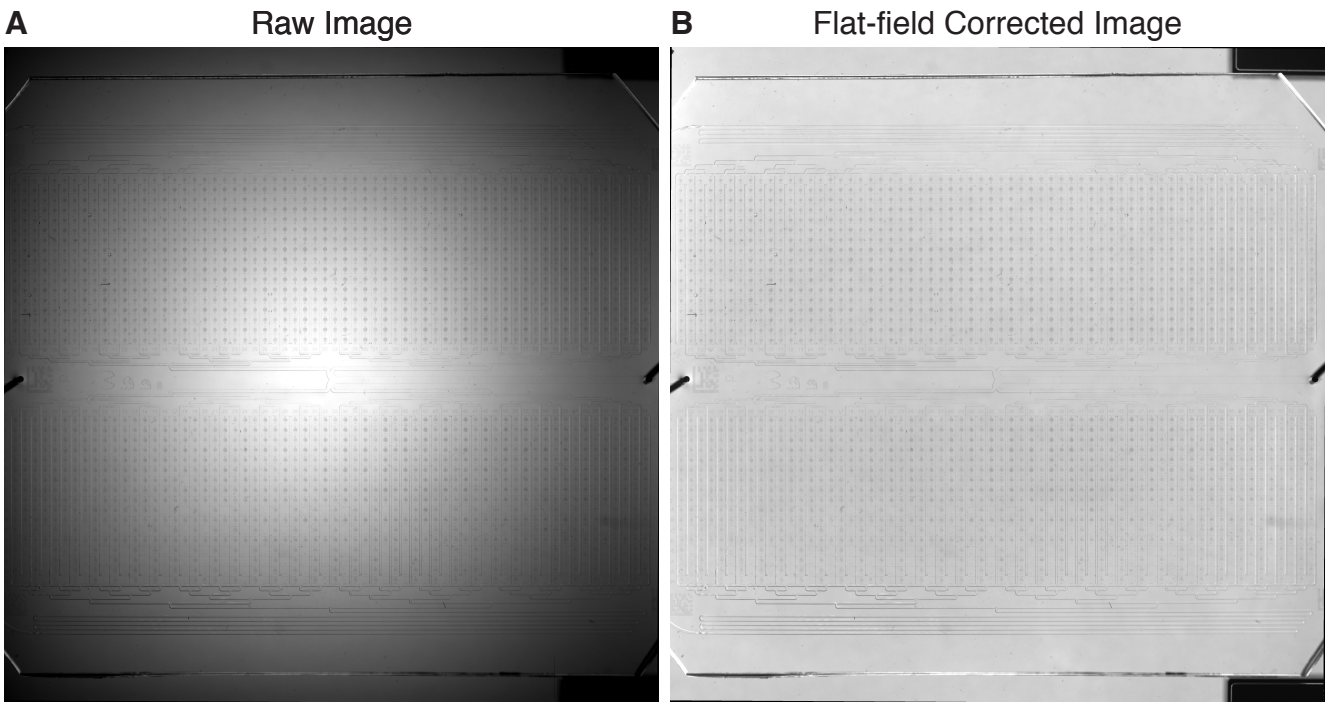

Fig. S2. A full Dynomics image taken on custom optics before (A) and after (B) flat-field correction.

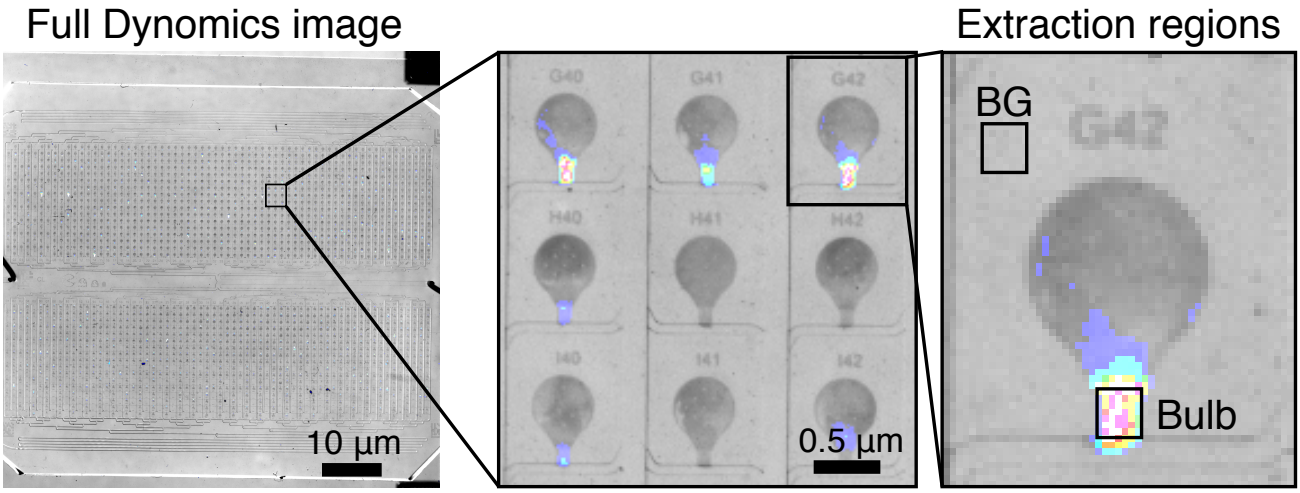

Fig. S3. Data extraction from a Dynomics image. 2,176 device positions are simultaneously imaged in both transmitted light (gray) and green fluorescence (16 color) channels. At each time point for each device position, the mean fluorescence of the boxed bulb and background (BG) regions are extracted.

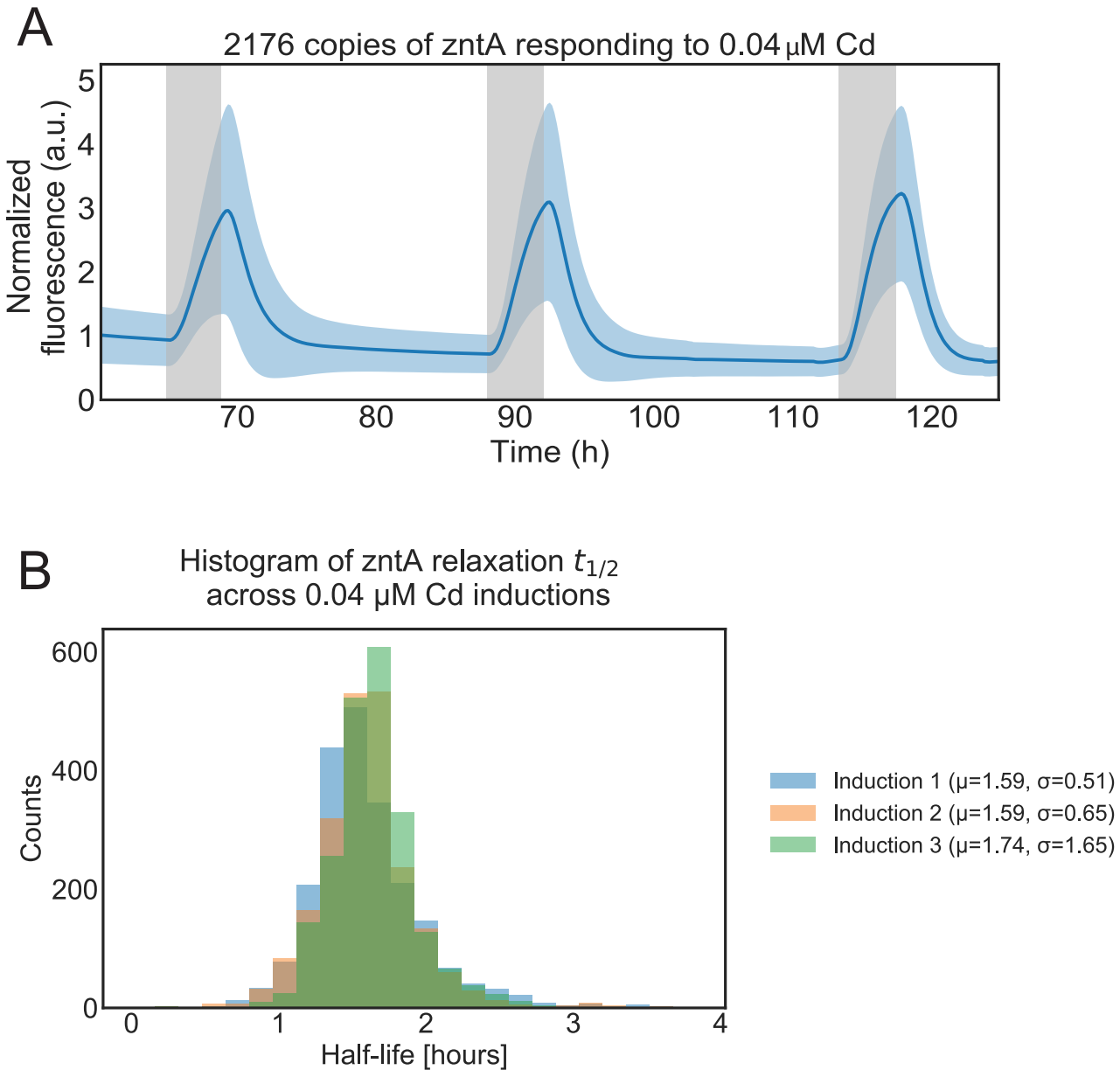

Fig. S4. Device validation experiment. To assay the variability of responses across our device, a Dynomics chip was spotted with 2176 replicates of the zntA strain from the *E. coli* GFP-promoter library (5). This strain is a reporter for the zinc and cadmium-responsive gene *zntA* and was chosen for its relevance in *E. coli* heavy metal metabolism and ability to be selectively activated via the heavy metal inducers used in the context of this work. (A) Mean (dark blue) plus/minus one standard deviation (light blue) for the 2716 strain responding to three consecutive four-hour 0.04 μM Cadmium inductions. (B) Histograms of the decay half-life of post-induction response (i.e the time taken for a strain's fluorescence levels to decay back down to their half-peak value) are overlaid for the three inductions plotted in (A).

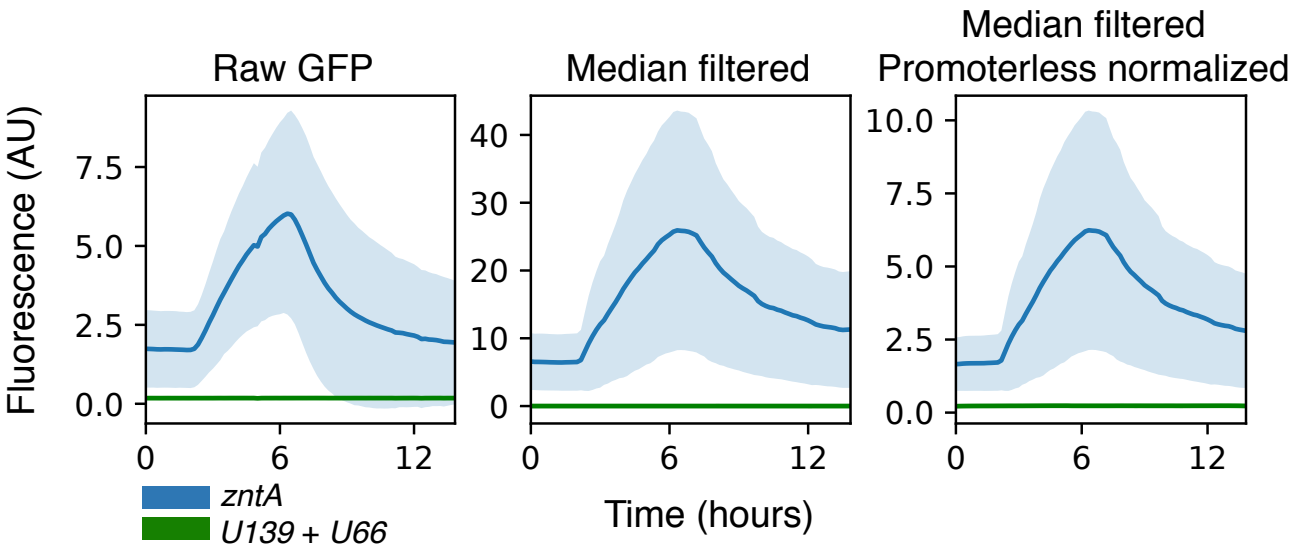

Fig. S5. Normalization process for data presented in Figure 2. Raw data was median filtered, then divided by the mean traces of promoterless strains U139 and U66. Dark lines show the mean of all time series while shaded regions represent the standard deviation.

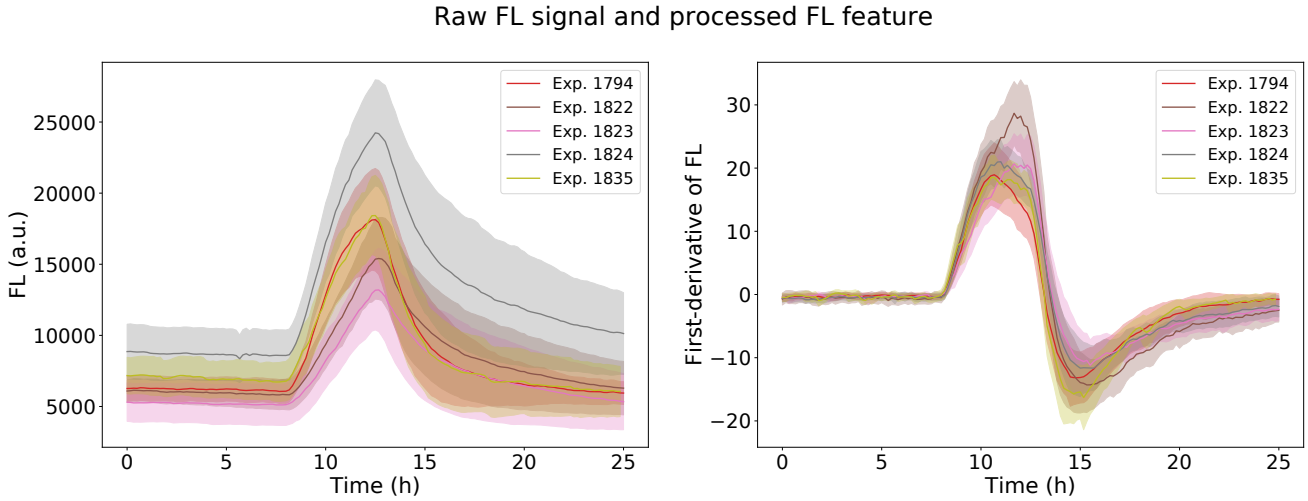

Fig. S6. Raw and processed fluorescent signals: Our feature engineering eliminates a significant amount of intra-experiment variability by rendering the raw signal into a first derivative-like feature. This variability is due to differing hardware between Dynomics devices, among other sources.

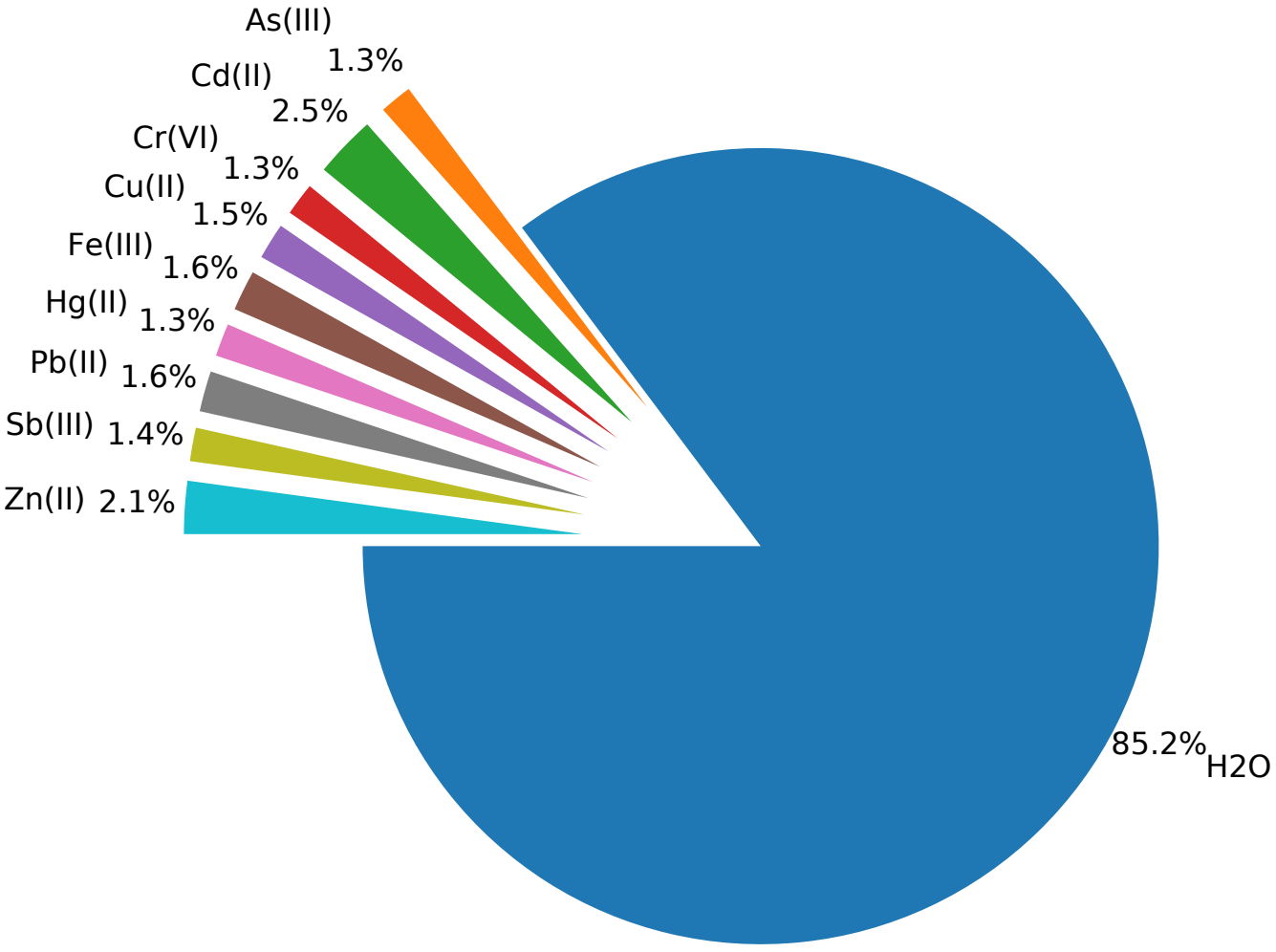

Fig. S7. The imbalanced class composition of our data set. This pie chart represents the composition of our data set by class. While 85.2% of the data set is composed of pure water, the remaining 14.8% is composed of a set of nine metals: lead, antimony, zinc, arsenic, cadmium, chromium, copper, iron and mercury. Of these metals, cadmium is the most frequent class, comprising 2.5% of the total data set.

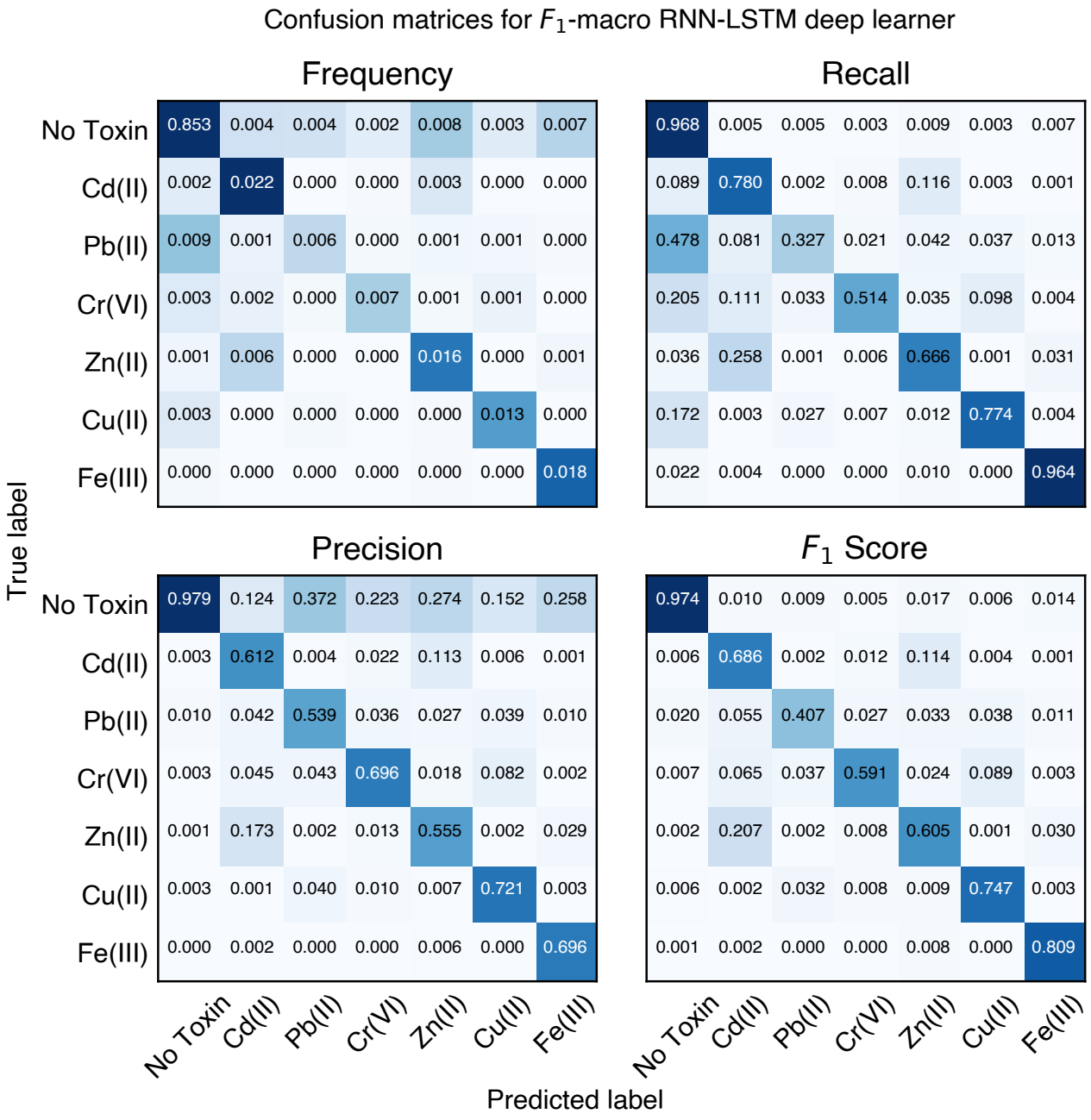

Fig. S8. The classification results for an LSTM-RNN. These confusion matrices represent the cross-validation results of our optimized LSTM-RNN classifiers. The recall and precision, as compared with those of the best-case XGBoost classifiers in Fig. S9, are dramatically improved. Although recall is not perfect, the LSTM-RNN rarely (at most 1% of the time) mistakes any metal for water when the metal is actually present, as can be observed in the left-hand column of the recall matrix. This result is an especially desirable one for a sensor: even if it is wrong about which metal is present, it is still detecting that something is contaminating the water. Similarly, the high rate of precision for water, at approximately 98%, is another desirable sensor quality, as it indicates a low false positive rate.

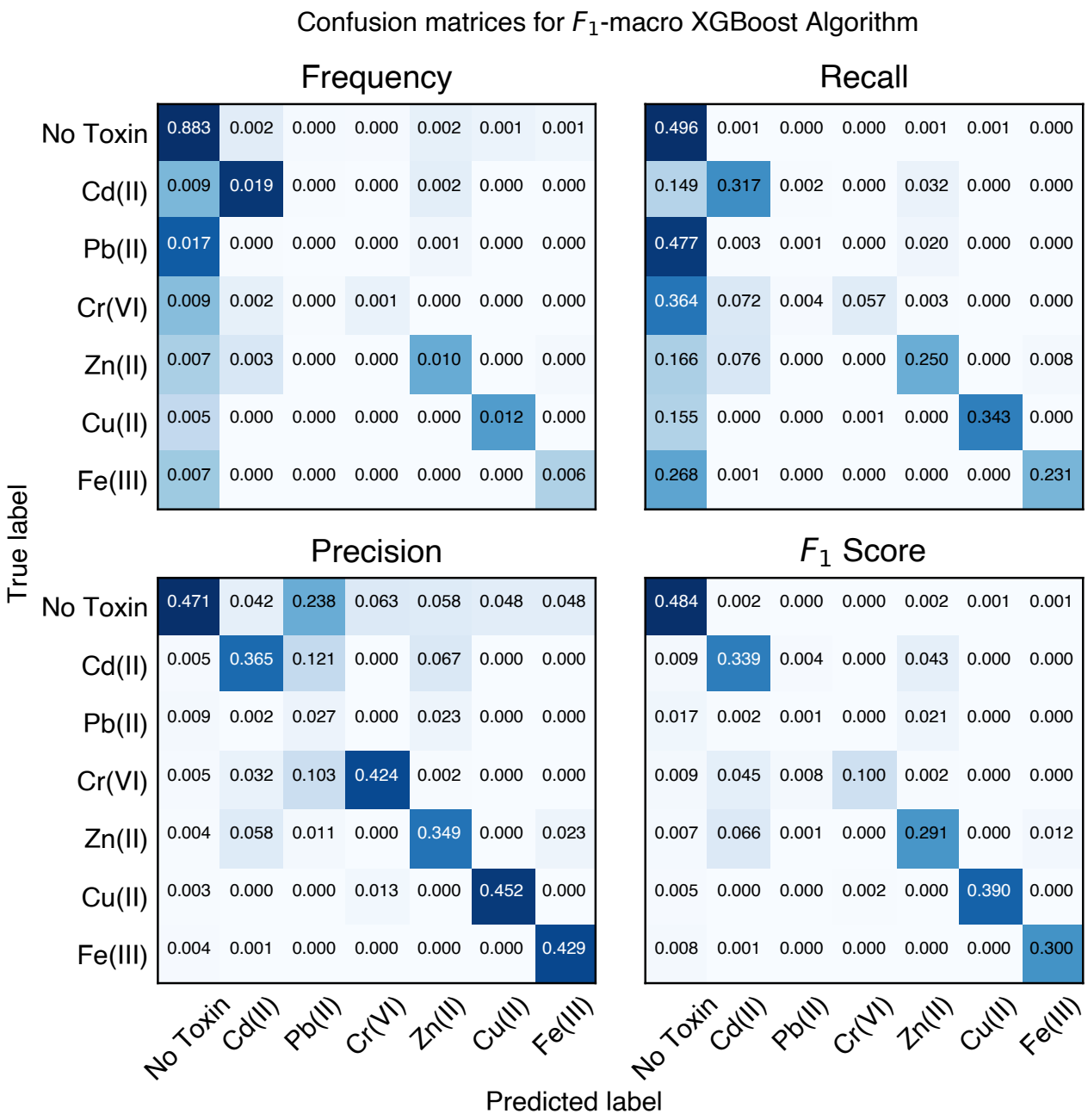

Fig. S9. The classification results for an XGBoost classifier. These confusion matrices represent the cross-validation results of our optimized XGBoost classifiers. While several metals are predicted with high precision and recall, this classifier does not match the performance of our best LSTM-RNN in Fig. S8.

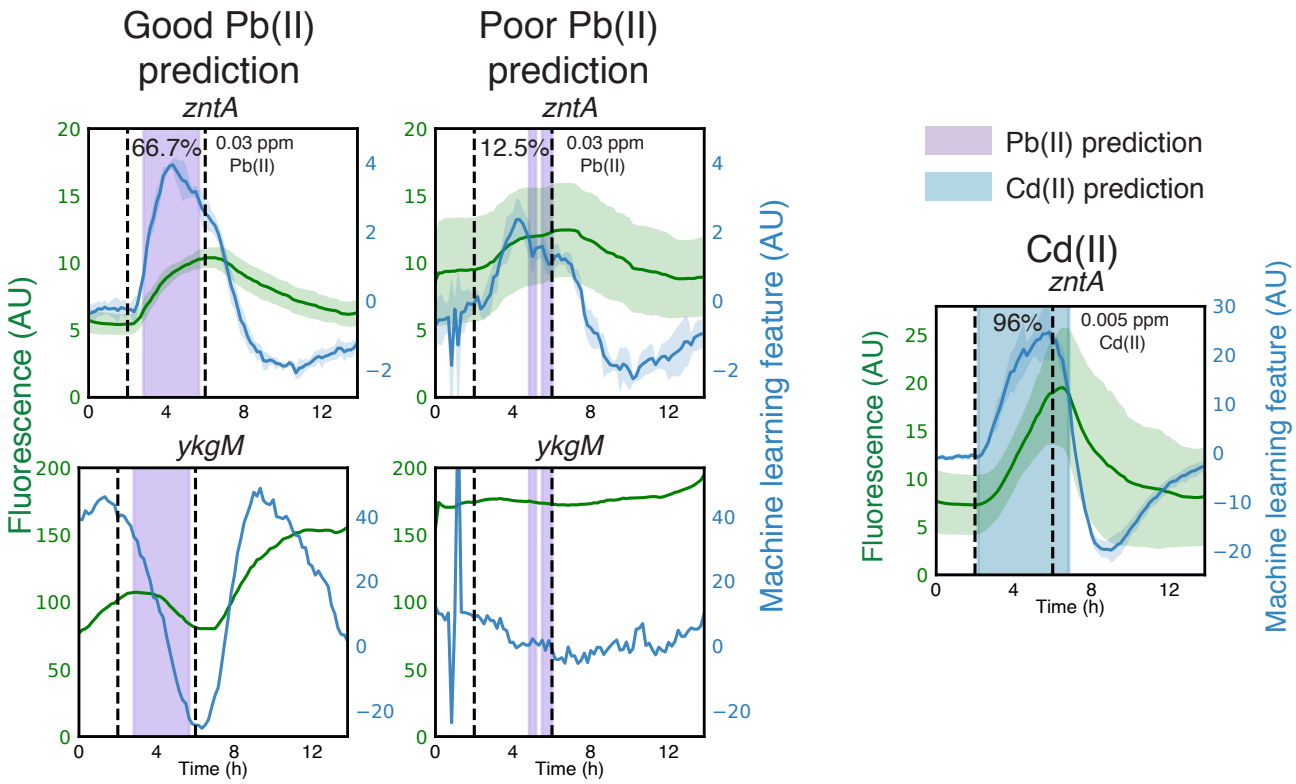

Fig. S10. A close look at lead classification. Two significant strains for lead detection as predicted by XAI, *zntA* and *ykgM*, were analyzed during the case of good and poor lead detection. Black dashed lines indicate metal exposure window, while color shaded region indicates the metal predicted by the LSTM classifier. At 0.03 ppm concentrations of Pb(II), fluorescence and the first-derivative like machine learning features are small as compared to those for a similar well-predicted metal, Cd(II). In both good (66.7% percent of Pb(II) time points correctly classified) and poor (12.5% correctly classified) prediction cases, Pb(II) is not predicted at the start of the induction window, though is predicted at some point during the 4 hour exposure. Solid traces indicate mean of replicate strains while shaded region is standard deviation. Only one copy of *ykgM* is present on each device.

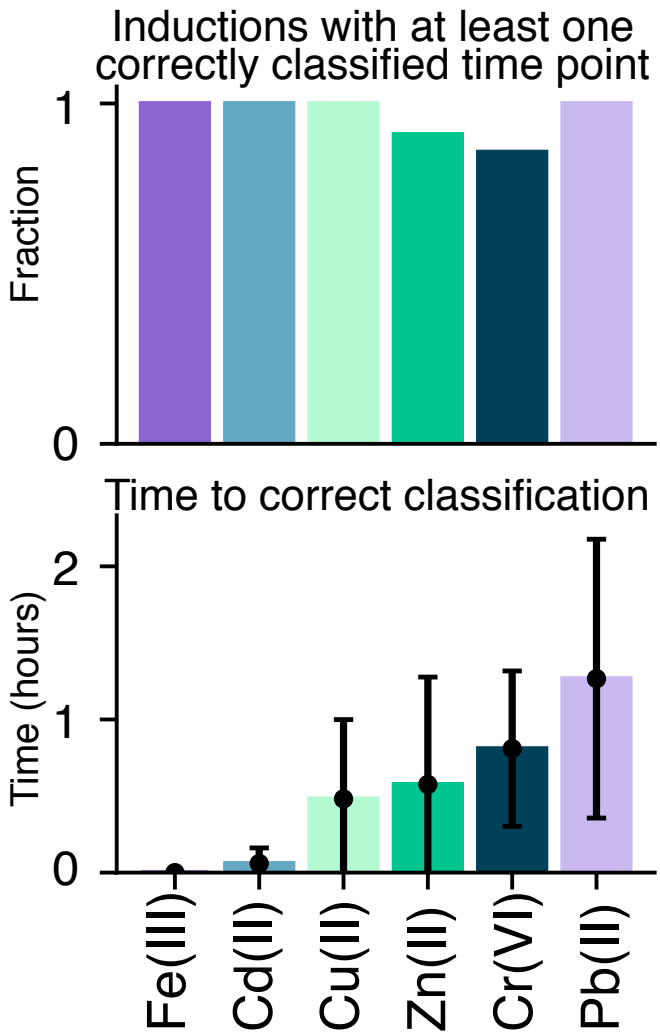

Fig. S11. A by-induction view of classifier performance. (Top) The fraction of 4 hour inductions that correctly classified at least one time point during the induction window. All inductions were correctly classified for all time points except for one of each chromium and zinc. (Bottom) The time of the first correctly classified time point during a four hour induction window. Metals with high recall scores in the confusion matrices are accurate predicted early within the induction window, while those with lower recalls take longer before the first accurately classified time point.

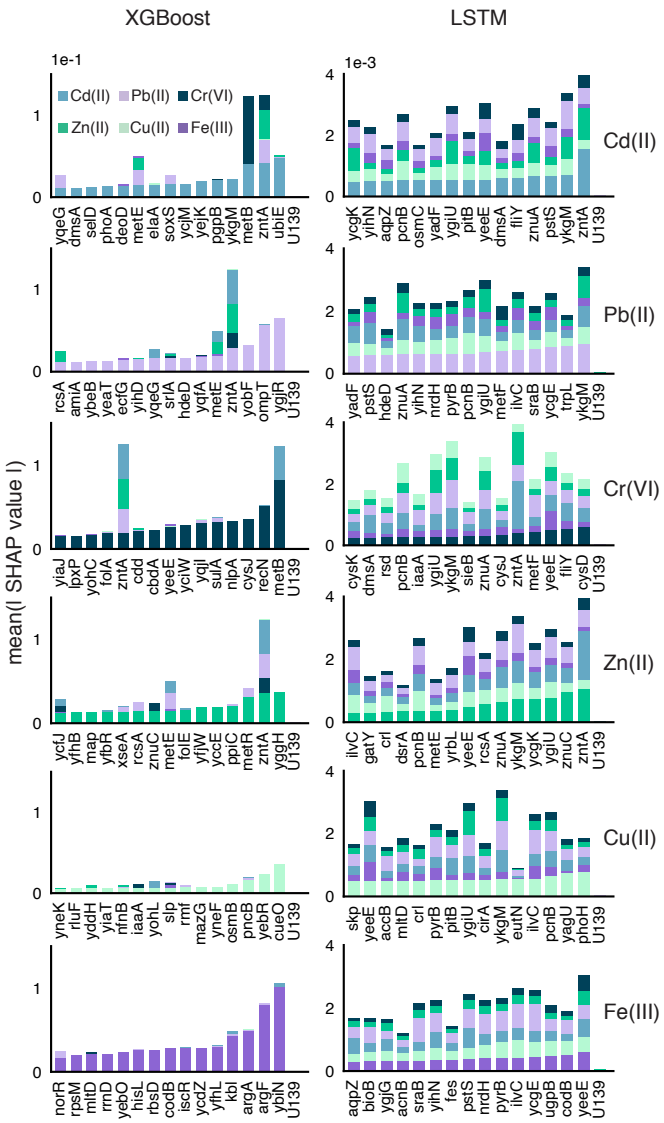

Fig. S12. Top 15 genes for the prediction of each metal as determined by XAI. The top 15 genes for each class were determined by taking the mean of the absolute SHAP values for each library member. The promoterless strain U139 is displayed as a negative control. We observed that XGBoost tends to identify unique strains for the prediction of each metal, while LSTM takes a more holistic approach.

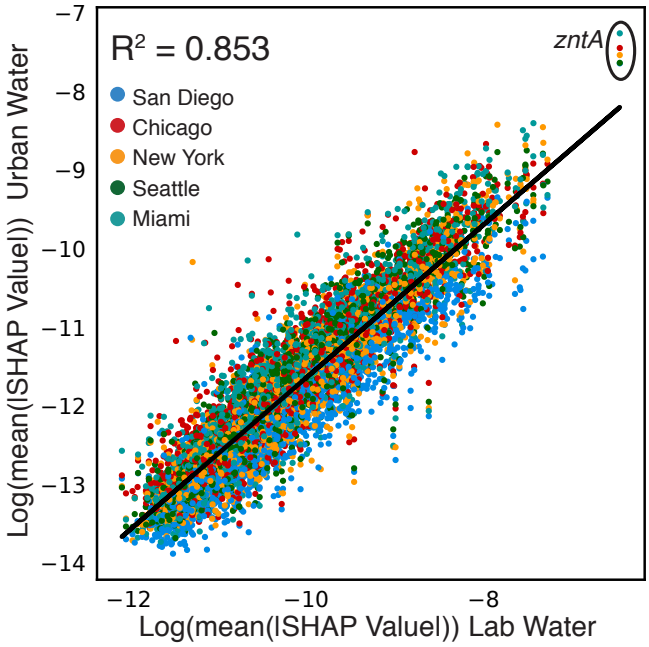

Fig. S13. The log mean absolute SHAP values for LSTM cadmium prediction for urban water and Milli-Q water plotted together. We observed that the *zntA* value is the best predictor of cadmium on all water sources. Other genes correlate between city and lab water sources.

Table S1. HM9 Minimal Media

Chemical	Vendor	Part No.	Concentration (mM)
Potassium Chloride	Millipore Sigma	P9541	49.6
MOPS pH 7.2	Millipore Sigma	PHG0007	40
β-Glycerol phosphate disodium salt pentahydrate	Millipore Sigma	50020	4
Dextrose	Millipore Sigma	D9434	22.2
Ammonium chloride	Millipore Sigma	326372	18.70
Magnesium Sulfate, 7-Hydrate	Macron Fine Chemicals	6066-04	0.2
Calcium Chloride Dihydrate	Fisher Scientific	C79-500	0.01
Iron(III) Chloride hexahydrate	Alfa Aesear	A16231	0.001
kanamycin sulfate from Streptomyces kanamyceticus	Millipore Sigma	K4000	0.086
Tween 20	bioWORLD	42030016-1	0.611
methyl α-D-mannopyranoside	Millipore Sigma	67770	50

Table S2. LSTM-RNN accuracy by metal with respect to water for the 18 standardized experiments. Each accuracy score reflects all instances where either the indicated metal or water were either present or predicted to be present. Accuracy was calculated by taking the ground truth and predictions for each metal, individually, performing a bitwise XOR operation on the resulting column vectors, then a bitwise NOT operation, summing the result, and dividing by the total number of timepoints.

Metal	Accuracy
Cadmium	98.4%
Chromium	99.4%
Copper	99.5%
Iron	99.3%
Lead	98.6%
Zinc	98.5%

Table S3. Concentration of metals in HM9 media made with San Juan River samples as measured by ICP-MS at the Environmental Complex Analysis Lab at UC San Diego.

Metal	Concentration
Aluminum	100.9 ppm
Vanadium	1.263 ppm
Chromium	88.83 ppb
Manganese	828.3 ppm
Iron	151.8 ppm
Cobalt	1.289 ppm
Nickel	1.634 ppm
Copper	1.215 ppm
Zinc	2.947 ppm
Gallium	63.73 ppb
Arsenic	64.94 ppb
Silver	22.45 ppb
Cadmium	48.16 ppb
Thallium	33.52 ppb
Lead	209.3 ppb

Table S4. Metal Induction Count by Concentration for Fold Change Analysis and Machine Learning Analysis

Metal	EPA MCL (ppb)	Concentration Tested (ppb)	Fold Change Induction Count	Machine Learning Induction Count
Arsenic	10	30	7	7
Antimony	6	1000	7	7
Cadmium	5	4.50	12	12
		8.99	0	1
Chromium	100	260	6	6
Copper	1300	127	8	8
Iron	300	279	6	6
		558	0	1
		5580	0	1
Lead	15	15	0	3
		30	6	6
		300	0	1
Mercury	2	2	6	6
Zinc	5000	50	5	5
		500	0	3
		1310	0	1

Table S5. Gene information for genes presented in manuscript Figures 2-4 1/2

Gene	Fig.	Protein names	Gene ontology (biological process)	Metal Binding
cueO	2	Blue copper oxidase CueO		Copper 1,2,3,4
aqpZ	2	Aquaporin Z	cellular water homeostasis , response to osmotic stress , water transport	
ycgE	2	HTH-type transcriptional repressor BluR		
ykgM	2	50S ribosomal protein L31 type B (Large ribosomal subunit protein bL31-B)		
codB	2	Cytosine permease	cytosine metabolic process , cytosine transport , proton transmembrane transport	
cysD	2	Sulfate adenylyltransferase subunit 2	hydrogen sulfide biosynthetic process , response to oxidative stress , sulfate assimilation	
trpL	2	Ubiquinone/menaquinone biosynthesis C-methyltransferase UbiE		
yeeE	2	UPF0394 inner membrane protein YeeE		
iscR	4	HTH-type transcriptional regulator IscR		(2Fe-2S)
acnB	4	Aconitate hydratase B	glyoxylate cycle , propionate catabolic process, 2-methylcitrate cycle , regulation of translation	(4Fe-4S)
ilvC	2,3	Ketol-acid reductoisomerase	branched-chain amino acid biosynthetic process , isoleucine biosynthetic process	Magnesium 1
zntA	2,3	Zinc/cadmium/lead-transporting P-type ATPase	cobalt ion transport , detoxification of zinc ion , lead ion transport	Zinc 1,2, Magnesium
fliY	2,3	L-cystine-binding protein FliY		
ycgK	2,3	Inhibitor of g-type lysozyme		
norR	4	Anaerobic nitric oxide reductase transcription regulator NorR		
sulA	4	Cell division inhibitor SulA	cellular response to DNA damage stimulus , division septum assembly	
iscR	4	HTH-type transcriptional regulator .		(2Fe-2S)
pstS	4	Phosphate-binding protein PstS	cellular response to DNA damage stimulus , phosphate ion transport	
yihN	4	Inner membrane protein YihN		
nrdH	4	Glutaredoxin-like protein NrdH		
ygiU	4	mRNA interferase toxin MqsR (Endoribonuclease MqsR)	negative regulation of transcription, negative regulation of translation , quorum sensing	
recN	4	DNA repair protein RecN	cellular response to DNA damage stimulus , double-strand break repair	
znuA	4	High-affinity zinc uptake system protein ZnuA		Zinc
pcnB	4	Poly(A) polymerase I	mRNA polyadenylation , mRNA processing , plasmid maintenance	
znuC	4	Zinc import ATP-binding protein ZnuC		
ygjG	4	Putrescine aminotransferase (Putrescine–2-oxoglutaric acid transaminase)		
ubiE	4	Ubiquinone/menaquinone biosynthesis C-methyltransferase UbiE	aerobic respiration , menaquinone biosynthetic process , ubiquinone biosynthetic process	
argF	4	Ornithine carbamoyltransferase subunit F	arginine biosynthetic process via ornithine , cellular response to DNA damage stimulus	
mltD	4	Membrane-bound lytic murein transglycosylase D	cell wall organization , peptidoglycan metabolic process , positive regulation of oxidoreductase activity	
cysJ	4	Sulfite reductase [NADPH] flavoprotein alpha-component	cysteine biosynthetic process , hydrogen sulfide biosynthetic process , sulfate assimilation	
metB	4	Cystathionine gamma-synthase	cysteine biosynthetic process via cystathionine , methionine biosynthetic process , transsulfuration	
pyrB	4	Aspartate carbamoyltransferase catalytic subunit	de novo' pyrimidine nucleobase biosynthetic process , 'de novo' UMP biosynthetic process	
pgpB	4	Phosphatidylglycerophosphatase B	glycerophospholipid biosynthetic process , peptidoglycan biosynthetic process	
metE	4	5-methyltetrahydropteroyltriglutamate–homocysteine methyltransferase	homocysteine metabolic process , methionine biosynthetic process , tetrahydrofolate interconversion	Zinc

Table S6. Gene information for genes presented in manuscript Figures 2-4 (continued)

Gene	Fig.	Protein names	Gene ontology (biological process)	Metal Binding
yadF	4	Carbonic anhydrase 2		Zinc
dmsA	4	Dimethyl sulfoxide reductase DmsA		(4Fe-4S),Molybdenum
bioB	4	Biotin synthase		(4Fe-4S)
argA	4	Amino-acid acetyltransferase		
argC	4	N-acetyl-gamma-glutamyl-phosphate reductase		
elaA	4	Protein ElaA		
hisL	4	his operon leader peptide (his operon attenuator peptide)		
ilvL	4	ilv operon leader peptide (ilvGMEDA operon attenuator peptide)		
infA	4	Translation initiation factor IF-1		
kbl	4	2-amino-3-ketobutyrate coenzyme A ligase		
metR	4	HTH-type transcriptional regulator MetR		
ompT	4	Protease 7 (Omptin) (Outer membrane protein 3B) (Protease A) (Protease VII)		
rbsD	4	D-ribose pyranase		
rpsM	4	30S ribosomal protein S13		
ugpB	4	sn-glycerol-3-phosphate-binding periplasmic protein UgpB		
ybiN	4	Ribosomal RNA large subunit methyltransferase F		
ycjM	4	Glucosylglycerate phosphorylase (GGa phosphorylase)		
yejK	4	Nucleoid-associated protein YejK		
yggH	4	tRNA (guanine-N(7)-)-methyltransferase		
ygjR	4	Uncharacterized oxidoreductase YgjR		
yqeG	4	Inner membrane transport protein YqeG		

Cluster	GO biological process complete	Fold Enrichment	P-value
1	localization (GO:0051179)	0.74	4.11E-02
	transport (GO:0006810)	0.71	3.30E-02
	transmembrane transport (GO:0055085)	0.62	4.47E-02
2	protein autoprocessing (GO:0016540)	9.7	4.59E-02
	transcription antitermination (GO:0031564)	9.7	4.59E-02
	carbon utilization (GO:0015976)	9.7	4.59E-02
	cellular response to light stimulus (GO:0071482)	9.7	4.59E-02
	cellular response to radiation (GO:0071478)	9.7	4.59E-02
	regulation of single-species biofilm formation (GO:1900190)	9.7	4.59E-02
	cellular response to UV (GO:0034644)	9.7	4.59E-02
	bacterial transcription (GO:0001121)	9.7	1.29E-02
	one-carbon metabolic process (GO:0006730)	7.28	2.11E-02
	tetrahydrofolate metabolic process (GO:0046653)	5.54	1.43E-02
	folic acid-containing compound metabolic process (GO:0006760)	4.31	2.68E-02
	pteridine-containing compound metabolic process (GO:0042558)	3.88	3.48E-02
	positive regulation of nitrogen compound metabolic process (GO:0051173)	2.11	3.44E-02
	macromolecule metabolic process (GO:0043170)	0.66	4.19E-02
	transmembrane transport (GO:0055085)	0.33	4.02E-02
3	thiamine-containing compound biosynthetic process (GO:0042724)	5.29	2.57E-02
	cellular detoxification (GO:1990748)	5.29	2.57E-02
	thiamine biosynthetic process (GO:0009228)	5.29	2.57E-02
	thiamine-containing compound metabolic process (GO:0042723)	4.23	4.07E-02
	thiamine metabolic process (GO:0006772)	4.23	4.07E-02
	antibiotic catabolic process (GO:0017001)	3.53	2.15E-02
	detoxification (GO:0098754)	3.53	2.15E-02
	cellular response to toxic substance (GO:0097237)	3.53	2.15E-02
	reactive oxygen species metabolic process (GO:0072593)	3.31	4.21E-02
	tetrapyrrole biosynthetic process (GO:0033014)	3.17	2.99E-02
	tetrapyrrole metabolic process (GO:0033013)	3.17	2.99E-02
	response to toxic substance (GO:0009636)	2.91	5.74E-03
	antibiotic metabolic process (GO:0016999)	2.8	1.46E-02
	vitamin biosynthetic process (GO:0009110)	2.65	1.29E-02
	water-soluble vitamin biosynthetic process (GO:0042364)	2.65	1.29E-02
	water-soluble vitamin metabolic process (GO:0006767)	2.43	1.78E-02
	vitamin metabolic process (GO:0006766)	2.43	1.78E-02
	drug catabolic process (GO:0042737)	2.41	2.54E-02
	drug metabolic process (GO:0017144)	2.17	1.33E-03
	positive regulation of biological process (GO:0048518)	0.38	4.72E-02
4	arginine biosynthetic process (GO:0006526)	2.79	4.34E-02
	arginine metabolic process (GO:0006525)	2.54	4.22E-02
	glutamine family amino acid biosynthetic process (GO:0009084)	2.36	3.63E-02
	drug transport (GO:0015893)	1.77	4.98E-02
	transmembrane transport (GO:0055085)	1.48	2.65E-02
	localization (GO:0051179)	1.28	4.48E-02
	drug metabolic process (GO:0017144)	0.5	2.90E-02

Fig. S14. Gene ontology enrichment analysis of the clusters obtained by agglomerative clustering of Fig 2A. Results were compiled from http://pantherdb.org/

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References

1. M. S. Ferry, I. A. Razinkov, and J. Hasty. *Microfluidics for synthetic biology: From design to execution*, volume 497. 2011. ISBN 9780123850751. .
2. Arthur Prindle, Phillip Samayoa, Ivan Razinkov, Tal Danino, Lev S. Tsimring, and Jeff Hasty. A sensing array of radically coupled genetic 'biopixels'. *Nature*, 2012. ISSN 00280836. .
3. Leslie A. Pratt and Roberto Kolter. Genetic analysis of Escherichia coli biofilm formation: Roles of flagella, motility, chemotaxis and type I pili. *Molecular Microbiology*, 30(2):285–293, 1998. ISSN 0950382X. .
4. Robert A. LaRossa, Dana R. Smulski, and Tina K. Van Dyk. Interaction of lead nitrate and cadmium chloride with Escherichia coli K-12 and Salmonella typhimurium global regulatory mutants. *Journal of Industrial Microbiology*, 14(3-4): 252–258, 1995. ISSN 01694146. .
5. Alon Zaslaver, Anat Bren, Michal Ronen, Shalev Itzkovitz, Ilya Kikoin, Seagull Shavit, Wolfram Liebermeister, Michael G. Surette, and Uri Alon. A comprehensive library of fluorescent transcriptional reporters for Escherichia coli. *Nature Methods*, 3(8):623–628, 2006. ISSN 15487091. . URL http://www.nature.com/doifinder/10.1038/nmeth895.
6. Huaiyu Mi, Anushya Muruganujan, Dustin Ebert, Xiaosong Huang, and Paul D Thomas. PANTHER version 14: more genomes, a new PANTHER GO-slim and improvements in enrichment analysis tools. *Nucleic acids research*, 47(D1): D419–D426, jan 2019. ISSN 1362-4962. . URL http://www.ncbi.nlm.nih.gov/pubmed/30407594http://www.pubmedcentral.nih.gov/articlerender.fcgi?artid=PMC6323939.
7. R.A. Rensink, J.K. O'Regan, and J.J. Clark. To See or Not to See: The Need for Attention to Perceive Changes in Scenes. *Psychological Science*, 8(5):368–373, 1997.
8. Joseph J. LaViola. Double exponential smoothing, 2004. URL https://www.itl.nist.gov/div898/handbook/pmc/section4/pmc433.htm.
9. Bobak Shahriari, Kevin Swersky, Ziyu Wang, Ryan P. Adams, and Nando De Freitas. Taking the human out of the loop: A review of Bayesian optimization. *Proceedings of the IEEE*, 104(1):148–175, 2016. ISSN 00189219. .
10. James Bergstra, Brent Komer, Chris Eliasmith, Dan Yamins, and David D. Cox. Hyperopt: A Python library for model selection and hyperparameter optimization. *Computational Science and Discovery*, 8(1), 2015. ISSN 17494699. .
11. Zachary Chase Lipton, Charles Elkan, and Balakrishnan Narayanaswamy. Thresholding Classifiers to Maximize F1 Score. *ArXiv e-prints*, 2014. URL http://arxiv.org/abs/1402.1892.
12. Martin Abadi, Paul Barham, Jianmin Chen, Zhifeng Chen, Andy Davis, Jeffrey Dean, Matthieu Devin, Sanjay Ghemawat, Geoffrey Irving, Michael Isard, Manjunath Kudlur, Josh Levenberg, Rajat Monga, Sherry Moore, Derek G Murray, Benoit Steiner, Paul Tucker, Vijay Vasudevan, Pete Warden, Martin Wicke, Yuan Yu, and Xiaoqiang Zheng. TensorFlow: A system for large-scale machine learning. In *12th USENIX Symposium on Operating Systems Design and Implementation (OSDI 16)*, pages 265–283, 2016. URL https://www.usenix.org/system/files/conference/osdi16/osdi16-abadi.pdf.
13. François Chollet and others. Keras. \url{https://keras.io}, 2015.
14. Mike Bernico. *Deep Learning Quick Reference*. O'Reilly, 2018. ISBN 9781788837996.
15. Rick van Hattem. Chapter 6: Generators and Coroutines - Infinity, One Step at a Time. In *Mastering Python*, chapter 6, pages 141–166. Packt Publishing, Birmingham, 2016.
16. Scott Lundberg. shap.
17. Scott M Lundberg, Gabriel Erion, Hugh Chen, Alex DeGrave, Jordan M Prutkin, Bala Nair, Ronit Katz, Jonathan Himmelfarb, Nisha Bansal, and Su-In Lee. Explainable AI for Trees: From Local Explanations to Global Understanding. *arXiv preprint arXiv:1905.04610*, 2019.
18. Scott Lundberg and Su-In Lee. A Unified Approach to Interpreting Model Predictions. (Section 2):1–10, 2017. URL http://arxiv.org/abs/1705.07874.
19. Avanti Shrikumar, Peyton Greenside, and Anshul Kundaje. Learning Important Features Through Propagating Activation Differences. 2017. URL http://arxiv.org/abs/1704.02685.
20. Alon Zaslaver, Avi E. Mayo, Revital Rosenberg, Pnina Bashkin, Hila Sberro, Miri Tsalyuk, Michael G. Surette, and Uri Alon. Just-in-time transcription program in metabolic pathways. *Nature Genetics*, 2004. ISSN 10614036. .
21. Laura S. Busenlehner, Mario A. Pennella, and David P. Giedroc. The SmtB/ArsR family of metalloregulatory transcriptional repressors: Structural insights into prokaryotic metal resistance. *FEMS Microbiology Reviews*, 27(2-3):131–143, 2003. ISSN 01686445. .
22. Tamar Barkay, Susan M. Miller, and Anne O. Summers. Bacterial mercury resistance from atoms to ecosystems. *FEMS Microbiology Reviews*, 27(2-3):355–384, 2003. ISSN 01686445. .
23. Joe J. Harrison, Howard Ceri, and Raymond J. Turner. Multimetal resistance and tolerance in microbial biofilms. *Nature Reviews Microbiology*, 5(12):928–938, 2007. ISSN 17401526. .